Do Analysts Understand Geopolitics? Evidence from Foreign Demand Shocks

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Abstract

Do sell-side analysts understand what geopolitical shocks do to firms' earnings? I measure their understanding by how much of a shock's predictable effect on a firm's earnings their forecast revisions incorporate, using 39 shocks of economic measures or war to 22 countries between 1995 and 2025. The predictable effect comes from where firms sell: the greater a U.S. firm's sales exposure to the shocked country, measured from its last 10-K before the shock, the further its earnings fall below analysts' pre-shock forecasts. Analysts' revisions incorporate only about a third of this predictable shortfall before the firm's next earnings announcement, and about two-thirds by their final forecast of the year's earnings. The under-reaction is not inattention, missed exposure, reluctance to deliver bad news or inexperience. Analysts revise in the right direction, for good news as well as bad, but too little, and their forecasts for exposed firms do not spread apart. They revise less after wars, whose consequences are hardest to gauge, than after economic measures. Analysts understand which firms a geopolitical shock hurts, but they underestimate how much it hurts them.

Keywords: geopolitical risk, sell-side analysts, earnings forecasts, under-reaction, foreign exposure, textual analysis

JEL codes: D84, F51, G14, G17, G24, M41

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1 Introduction

Wars, sanctions and tariffs have become a recurring source of risk for U.S. firms. In a 2025 survey of more than 1,300 experts, the two risks most often named as the most likely to trigger a global crisis in 2026 were both geopolitical, namely economic confrontation between states and armed conflict (World Economic Forum, 2026). Geopolitical risk is also costly, lowering investment and employment (Caldara and Iacoviello, 2022). Sanctions and tariffs, which governments increasingly use as instruments of power (Mohr and Trebesch, 2025; Clayton et al., 2026), reach ordinary firms through the countries the firms trade with. U.S. firms that traded with China, for example, lost value when the 2018 tariffs were announced (Huang et al., 2023). War spreads the same way. Stock markets closer to Ukraine fell more when Russia invaded in 2022, largely through trade links (Federle, Meier, Müller, and Sehn, 2026).

Translating such an event into a firm's earnings is difficult, and investors rely on sell-side analysts to do it (Brown et al., 2015; Kothari et al., 2016). A geopolitical shock originates in foreign politics, varies in form from one episode to the next (Table 1), and has consequences that depend on how governments respond (Crozet and Hinz, 2020; Fajgelbaum et al., 2020), how long the measures last and how currencies move (Bartov and Bodnar, 1994). Analysts' expertise is built around industries and firms (Boni and Womack, 2006; Bradley et al., 2017; Mikhail et al., 1997). Their brokerages employ economists who follow such events. Analysts whose brokerage has an active in-house macroeconomist respond more fully to bad macroeconomic news (Hugon et al., 2016), and analysts forecast better than managers when macroeconomic conditions drive earnings (Hutton et al., 2012). Yet the translation into the earnings of a particular firm falls to the analyst who covers it, and it requires knowing where the firm sells, buys and produces and how each of these ties responds. If analysts make the translation slowly, prices that depend on their forecasts may adjust slowly as well.

This paper asks how well analysts make this translation, and it measures how well by the forecasts themselves. When a geopolitical shock hits a country, how much of the shock's effect on the earnings of U.S. firms that sell there do analysts' revisions incorporate? An analyst who understands the shock revises by its whole expected effect, a share of one, and an analyst who ignores it does not revise, a share of zero. I ask how large this share is, how quickly it grows and, if it falls short of one, why.

Existing evidence does not settle the question. Analysts adjust only partly for expected inflation (Basu et al., 2010), under-react to bad news about GDP growth (Hugon et al., 2016) and to an index of quarterly macroeconomic news (Dai et al., 2025), and do not fully resolve the uncertainty that interest-rate, exchange-rate and commodity-price shocks create

for exposed firms' earnings (Guay et al., 2003). Such news recurs every quarter, however, so analysts can learn its effects on a firm from many past quarters. A geopolitical shock is rare, varies in form and reaches a firm through its ties to one foreign country, which the analyst must first identify. Its prominence could make analysts respond more fully than to routine news, or its novelty could make them respond less. These studies show that forecasts leave errors that recurring news predicts. They do not isolate a dated, public event whose consequences for each firm are predictable from its own filings, or follow how much of those consequences analysts incorporate before and after the firm reports.

Measuring how much analysts incorporate is hard with the usual tools, because a firm's ties to a country can push its earnings in opposite directions (Guidolin and La Ferrara, 2007). A sanction that closes a market hurts the firms that sell there, while the currency depreciation that often follows lowers the costs of firms that produce there. Studies that relate analysts' forecasts to an aggregate risk index (Chourou et al., 2021; Zhu et al., 2026) therefore cannot say how much a given shock should move a given firm's earnings, or in which direction. This requires a measure of each firm's exposure fixed before the shock and a benchmark for the revision that the shock warrants.

I focus on foreign demand, the one channel whose effect has a predictable sign. A shock that closes a country's market, weakens its currency or reduces its demand is a foreign demand shock in the sense of the trade literature (Hummels et al., 2014), and firms that sell in the country should, on average, earn less after it. Firms that buy or produce there, by contrast, face offsetting forces. Following Hoberg and Moon (2017), I measure sales exposure as mentions of the shocked country near sales words in each firm's latest 10-K before the shock, and I hold fixed sourcing and operations exposure, measured the same way near sourcing and operations words.

Within an industry and a shock, sales exposure should therefore predict both how far a firm's actual earnings fall short of an analyst's pre-shock forecast and how far the analyst revises that forecast after the shock. Dividing the second by the first gives the incorporation rate, the share of the exposure-driven forecast error (the part of the pre-shock forecast error that sales exposure predicts) that analysts' revisions incorporate. It is an instrumental-variable estimate with sales exposure as the instrument, and I call this approach the signed design, since it relies on the direction of the effect. Because both the 10-K and the shock are public, efficient forecasts would incorporate all of the exposure-driven forecast error on average across shocks, so a rate below one measures under-reaction. The rate maps directly into the forecast-error-on-revision regression of Coibion and Gorodnichenko (2015), so it can be compared with other studies of forecasters (Section 4.1).

The shocks come from the country-level geopolitical risk indexes of Caldara and Iacoviello

(2022) and Caldara et al. (2026). From these indexes I select 39 shocks, episodes of sharply elevated risk to 22 countries between 1995 and 2025 that involve economic measures against the shocked country or war on or against its territory (Table 1 and Figure 1). They arise from 26 events, since one conflict or crisis can produce shocks in several countries or years. In the main sample of 29,177 firm-shock and 268,865 analyst-firm-shock observations, I follow each analyst's forecast of street earnings per share (EPS) from the pre-shock forecast, the last before the shock, through three post-shock forecasts. The pre-EA forecast is the one in effect just before the firm's first earnings announcement (EA) after the shock's peak, the post-EA forecast is the first forecast on or after that announcement, and the final forecast is the last before the target year's annual earnings announcement (Figure 2).

Sales-exposed firms, those whose 10-K mentions selling to the shocked country, fall short of their pre-shock forecasts. Within the same industry and shock, the shortfall of the median sales-exposed firm exceeds that of a firm without sales exposure by about 1.7 percent of annual EPS, and at the 90th percentile of sales-exposed firms by about 5 percent (Section 5.2). Analysts revise in the right direction, but by too little. By the pre-EA forecast, their revisions incorporate 0.36 $[0.17, 0.79]$ of the exposure-driven forecast error (brackets give 95 percent confidence sets robust to weak instruments). The set excludes both 0 and 1, and rejecting 1 does not rely on the exclusion restriction, because efficient forecasts would leave no error that public information, sales exposure included, could predict (Section 4.2). Analysts thus understand which firms a geopolitical shock hurts, but they underestimate how much it hurts them. Over the rest of the year, analysts' own forecasts reach 0.44 $[0.17, 0.99]$ by the post-EA forecast and 0.69 $[0.43, 0.93]$ by the final forecast (Table 4 and Figure 4), mostly after the firm's first announcement rather than at it. The consensus forecast, which drops estimates that analysts stop updating, reaches 0.92 $[0.81, 1.11]$ by the end of the year (Table 5), so forecasts catch up.

Several checks show that the estimates come from the shock and from where firms sell, rather than from standing differences between firms, the assumed direction of the effect, what else the 10-K says about the country, or chance. The same firms show no comparable forecast errors or revisions in the two years before the shock (Table 5), and analysts' revisions before the shock started are small and not significant (Appendix Figure A3). Replacing exposure to the shocked country with exposure to each of 20 other countries produces nothing as strong in the same direction (Table 6), and an unsigned design that does not assume the direction of the effect gives a pre-EA rate of 0.31 $[0.05, 0.93]$ (Table 7). Counted one section at a time, sales exposure predicts most strongly in Management's Discussion and Analysis (MD&A), where managers must discuss known trends, and not at all in the risk factors (Table 8), and mentions of the shocked country without sales, sourcing or operations words predict nothing

in the signed design. Wild cluster bootstrap and randomization inference give much the same p-values as clustered standard errors, and leaving out any one shock keeps the pre-EA rate between 0.30 and 0.44 (Table 6 and Appendix Table A8).

Why, then, do analysts under-react? Prior research suggests several reasons. They may not attend to news outside their usual focus (Hirshleifer and Teoh, 2003; Driskill et al., 2020). They may overlook a firm's exposure in a long filing, much as investors overlook changes in 10-K text (Cohen et al., 2020). They may be reluctant to deliver bad news, because optimism serves their careers and their access to management (Lim, 2001; Hong and Kubik, 2003). They may lack experience of such shocks, as experience of earlier epidemics helped analysts during the COVID-19 pandemic (Agarwal and Chalwati, 2023). Or they may notice the shock and the exposure but adjust too little, the sluggish updating found among professional forecasters and analysts (Coibion and Gorodnichenko, 2015; Bouchaud et al., 2019), which may be stronger when the consequences are hard to gauge (Zhang, 2006). Geopolitical shocks make the last possibility plausible. Their direction for firms that sell in the shocked country is clear, but the size of their effect depends on how the episode unfolds. The setting can tell these explanations apart, because each of the first four names a group in which the under-reaction should be much smaller, namely analysts who follow the shock, firms whose exposure is easy to see, good news, and analysts with experience of an earlier episode in the same country.

In none of these groups is the under-reaction much smaller. Analysts notice the shock. They revise the forecasts of sales-exposed firms sooner than those of other firms, forecasts not yet revised explain little of the under-reaction, and analysts who were following the shock incorporate no more than colleagues covering the same firm (Section 7.1). They do not miss the exposure, since they incorporate no more when it is visible as a geographic segment named for the shocked country (Section 7.2). Nor are they reluctant to deliver bad news (Easterwood and Nutt, 1999). Firms that operate in the shocked country beat their pre-shock forecasts, and analysts incorporate that good news no faster than the bad news, at rates of 0.29 $[0.05, 0.50]$ and 0.36 that cannot be told apart (Table 9). And neither experience of an earlier episode in the same country nor working at one of the largest brokerages, which are the most likely to employ economists, raises how much analysts incorporate before the firm reports (Table 10).

What remains is a shared under-adjustment: analysts revise as if the shock's expected effect on earnings were smaller than it is. They move in the right direction but too little, and they move together, since their forecasts for sales-exposed firms do not spread apart after the shock. They also revise less after wars, whose economic consequences are hardest to gauge, than after economic measures, although sales exposure predicts a shortfall of similar

size (Section 7.5).

The paper contributes to three literatures. First, it adds to recent work on how analysts respond to geopolitical risk. Ji et al. (2026) study one shock, the 2018 U.S.–China trade war, with exposure inferred from firms' past sensitivity to geopolitical risk. Zhu et al. (2026) relate forecast errors to an aggregate geopolitical risk index, and Cuculiza et al. (2021) show that analysts located near terrorist attacks turn pessimistic. By contrast, I study 39 shocks to 22 countries, measure each firm's exposure from its 10-K before the shock, compare firms within an industry and a shock, and estimate how much of a shock's predictable effect analysts incorporate and when.

Second, the paper relates to work on how analysts and investors process firms' foreign exposure. Analysts under-react to expected growth in the countries where firms sell (Ma et al., 2016), past currency changes predict the forecast errors of firms with foreign operations (Bartov and Bodnar, 1994), and investors under-react to news from the countries where firms operate (Huang, 2015). The setting here differs in three ways. The news is a discrete, public geopolitical event rather than gradual macroeconomic change. Sales exposure is separated from sourcing and operations exposure, and the good news that operations exposure brings is incorporated no faster than the bad. Finally, the exposure measure, which builds on text-based measures of firms' foreign and political exposure (Hoberg and Moon, 2017; Hassan et al., 2019, 2024), is fixed before the shock and public when analysts forecast. It captures where firms sell rather than general ties to a country, and it predicts most strongly when counted in the MD&A, a section in which managers must discuss known trends (U.S. Securities and Exchange Commission, 2021) and which the literature finds informative (Bryan, 1997; Li, 2010). Analysts' revisions do not respond to changes in MD&A text (Brown and Tucker, 2011), and markets miss changes in 10-K text (Cohen et al., 2020). Here, by contrast, finding the exposure is not the problem, since analysts incorporate no more when it is easy to see. What they get wrong is how far to adjust.

Third, the paper contributes to the study of how forecasters form expectations and why analysts' forecasts err. Evidence of under-reaction in professional forecasts (Coibion and Gorodnichenko, 2015) and in analysts' firm-level forecasts (Abarbanell and Bernard, 1992; Bouchaud et al., 2019) usually relates forecast errors to past revisions or past earnings. Other studies follow forecasts after a dated event, such as macroeconomic forecasts after Russia's invasion of Ukraine (Gorodnichenko and Vasudevan, 2026) and analysts' forecasts after temperature shocks (Addoum et al., 2023) or during the COVID-19 pandemic (Landier and Thesmar, 2020). I isolate the predictable consequences of a dated, public event for each firm, measure how much of them analysts incorporate, and trace the incorporation rate over the year. A geopolitical shock also falls between the cases that studies of over-

and under-reaction contrast, since it is common to a country but reaches each firm through its own exposure. Forecasters appear to under-react to aggregate shocks but over-react to idiosyncratic ones (Kučinskas and Peters, 2024), analysts over-react in their long-term growth forecasts (Bordalo et al., 2019), and in macroeconomic surveys individual forecasters over-react to news while the consensus under-reacts (Bordalo et al., 2020). Here analysts under-react in their own forecasts, not only in the consensus, and their forecasts do not spread apart, as they would if some analysts incorporated much more than others. Of the explanations for analysts' forecast errors, the evidence favors a shared under-adjustment over incentives to stay optimistic, inattention or inexperience. Experience of an earlier episode in the same country does not speed analysts' response before the firm reports, which adds to work on analysts' experience (Mikhail et al., 2003; Agarwal and Chalwati, 2023) and on how experience shapes expectations (Malmendier and Nagel, 2011).

The rest of the paper is organized as follows. Section 2 describes the shocks, how they reach firms' earnings and the hypotheses, Section 3 the data and measurement, and Section 4 the research design. Section 5 reports the main results, Section 6 their robustness, including the unsigned design, Section 7 asks why analysts under-react, and Section 8 concludes.

2 Setting and Hypotheses

Asking whether analysts incorporate what a geopolitical shock does to a firm's earnings requires shocks that are large, dated and public, as well as a reason to expect a particular effect on particular firms. This section describes the shocks, explains why a shock should lower the earnings of U.S. firms that sell in the shocked country while its effect on firms that buy or produce there has no clear sign, and states the hypotheses.

2.1 Geopolitical shocks

The geopolitical shocks come from the country-level geopolitical risk indexes of Caldara and Iacoviello (2022), extended to 44 countries by Caldara et al. (2026), which count newspaper articles that discuss geopolitical risk and mention a given country.[1] The indexes follow the newspaper-count approach of Baker et al. (2016). Rather than the index itself, I use each country's share of geopolitical-risk news, the percent of geopolitical-risk articles that mention it. A country's index also rises when geopolitical news surges everywhere, whereas its share rises only when the country becomes a larger part of that news (Appendix A.2). Following

[1]Following Caldara and Iacoviello (2026), I use "country" for every place their index covers, including Hong Kong and Taiwan.

the threshold logic of Bloom (2009), I define an episode as a run of months in which a country's share exceeds its trailing five-year norm by more than two standard deviations. The run gives each episode a first month, a peak month and a last month.

Most such episodes, however, are not economic events. The index measures attention to geopolitical events rather than their economic stakes. Of the 377 episodes in which a country is directly involved, for example, 146 are mainly threats and 49 mainly terrorist attacks, and only 102 involve economic measures or war at all (Appendix Table A3). I therefore keep an episode only if it peaks at least three standard deviations above the norm, if the country is on the receiving end of economic measures (sanctions, tariffs, embargoes or asset freezes) or of war on or against its territory, and if the qualifying measure or war is public by the end of the peak month.

The thresholds and the timing rule each do a specific job. Dating episodes at two standard deviations places the pre-shock forecast before the build-up that precedes many shocks, while selecting them at three standard deviations keeps only the most prominent shocks, which are the most likely to move earnings. Requiring the measure or war to be public by the peak month ensures that analysts could have known about it when they issued their pre-EA forecasts. To apply these rules, I hand-code each episode's type and the country's role in it, checking every label against its sources, and an independent check traced every qualifying measure or war to a dated source.

Table 1 lists the 39 shocks that result, from 26 events between 1995 and 2025, and Figure 1 places them in time. Of these, 33 begin after 2000. They include the sanctions on Russia in 2014 and 2022, the U.S.–China trade war and the steel and aluminum tariffs of 2018, the 2025 tariffs on Mexico and Canada, and the wars in Ukraine and Israel. Twenty-four shocks involve economic measures only, eight involve war only, and seven involve both. Section 7.5 compares the 24 shocks without war with the 15 that involve war. An event is the conflict or crisis behind a shock, named alike in every country, and one event can produce several shocks. The Russo-Ukrainian War produced shocks to Russia, Ukraine, Germany and Italy in 2022, and long-running conflicts, such as those in Colombia and Mexico, produced repeated shocks in the same country. Because shocks at the same time can share news, Section 6.5 also leaves out and clusters by groups of overlapping shocks.

2.2 How shocks reach firms' earnings

A shock to a foreign country reaches a U.S. firm's earnings through the firm's ties to that country, that is, through where it sells, where it buys and where it operates. These ties do not all push in the same direction.

For firms that sell in the shocked country, the main forces point the same way. War and sanctions reduce output and demand in the country that suffers them (Neuenkirch and Neumeier, 2015; Federle, Meier, Müller, Mutschler, and Schularick, 2026). A falling local currency shrinks the dollar value of revenue earned there (Bartov and Bodnar, 1994), and sanctions, counter-sanctions and retaliatory tariffs block sales directly (Crozet and Hinz, 2020; Fajgelbaum et al., 2020). Consistent with these forces, firms that sold to countries in crisis lost value when the crises hit (Forbes, 2004).

For firms that buy from the shocked country, by contrast, the forces conflict. Disruptions at suppliers reduce their customers' output (Barrot and Sauvagnat, 2016; Boehm et al., 2019), and tariffs raise the cost of imported inputs (Amiti et al., 2019). Yet a falling currency can make inputs cheaper, firms reroute purchases to other countries (Freund et al., 2024), and they pass part of higher costs on to their customers (Cavallo et al., 2021). The net effect of a shock on a firm that sources from the shocked country has no clear sign. Selling abroad and buying inputs abroad carry different risks (Hoberg and Moon, 2019).

The same is true of firms that operate there. Conflict can hurt firms with business in the affected region (Abadie and Gardeazabal, 2003) or help them (Guidolin and La Ferrara, 2007). A falling currency lowers local costs, less so for firms that import their inputs (Amiti et al., 2014), and plants that export from the country can gain.

Selling is therefore the only tie for which theory predicts the sign of a shock's effect on earnings. This is why I measure exposure through sales and hold sourcing and operations exposure fixed. A measure that counts all ties to a country mixes effects of opposite sign, and those effects can cancel. The effect of selling need not be negative in every shock or for every firm. The prediction concerns its average across shocks, relative to other firms in the same industry and shock, holding sourcing and operations ties fixed, and it is the first of the hypotheses that follow.

2.3 Hypotheses

If selling to the shocked country lowers earnings on average, sales-exposed firms should fall short of analysts' pre-shock forecasts by more than other firms in the same industry. Their pre-shock forecast error, actual EPS minus the forecast issued before the shock, should then decrease with sales exposure (H1). If analysts react to the shock, they should also revise their forecasts for these firms downward, so that the forecast revision, the change from the pre-shock forecast to a later one, decreases with sales exposure (H2). Section 4.1 defines both variables formally and states each hypothesis in terms of the estimating equations.

The third hypothesis concerns how far the revisions go. The part of the pre-shock forecast

error that sales exposure predicts is the exposure-driven forecast error, and the incorporation rate is the share of it that analysts' revisions incorporate. Sales exposure is public in the 10-K, and the shock is public by its peak. If forecasts used all public information, the forecast error left after the revision would not vary with sales exposure on average across shocks, and the incorporation rate would equal one (Mincer and Zarnowitz, 1969; Nordhaus, 1987). This efficiency benchmark is H3. Under-reaction instead gives a rate between zero and one, and over-reaction a rate above one. The benchmark does not require the shock's consequences to be certain. An analyst who is unsure how much a shock will lower a firm's earnings should still move the forecast by the expected effect, and the error that remains would then be unrelated to sales exposure. A rate below one therefore means that forecasts stay predictably too high for sales-exposed firms, whatever analysts' reasons for revising less.

Under-reaction would be consistent with evidence that analysts (Mendenhall, 1991; Abarbanell and Bernard, 1992) and other professional forecasters (Coibion and Gorodnichenko, 2012, 2015) underweight new information, analysts especially negative information (Easterwood and Nutt, 1999). The closest evidence concerns macroeconomic news. Analysts only partly adjust their forecasts for expected inflation (Basu et al., 2010), under-react to bad macroeconomic news (Hugon et al., 2016), and under-react to quarterly macroeconomic news more when macroeconomic uncertainty is high (Dai et al., 2025).

3 Data and Measurement

The analysis combines three sources. I measure each firm's ties to the shocked country from the text of its 10-K, take analysts' earnings forecasts from I/B/E/S, and take firm data from CRSP, Compustat and Audit Analytics. This section describes the measure of sales exposure, the forecasts and the sample.

3.1 Measuring sales exposure

Exposure is measured from the firm's own annual report. For each firm and shock, the *10-K in use* is the firm's latest 10-K filed on EDGAR before the shock's first month, if it is at most 18 months old. While an episode of elevated geopolitical risk (any episode of the index, not only the 39 shocks) is under way in a country that the 10-K mentions, the 10-K in use is not replaced by a newer filing. Exposure is therefore always measured before the shock and never reflects the firm's own response to it. A 10-K kept in use this way can be older than 18 months, which happens in 2 percent of firm-months (Appendix A.3). Because the 10-K in use is public when the shock begins, analysts can read it. They do read filings, and EDGAR

use by their brokerages accompanies more accurate forecasts (Gibbons et al., 2021).

Within the 10-K in use, I count mentions of the shocked country, following studies that measure where firms do business from place names in their filings (García and Norli, 2012) or earnings calls (Hassan et al., 2024). A rule-based country dictionary identifies each mention. An audit of the dictionary's 200 most frequent terms, on random samples of 10-K passages, removes terms that mostly refer to something else, such as "Jordan" as a person's name or "Jersey" in "Jersey City", and blocks phrases in which a country's name does not refer to the country.

Each mention is then sorted by what it is about. Following Hoberg and Moon (2017), a mention belongs to a group of business words when one of the group's words appears within 25 words of it in the same paragraph. Sales words include sales, revenue, customers, markets, distribution, exports and demand. Sourcing words include suppliers, imports, vendors and "purchase ... from", and operations words include subsidiaries, facilities, plants, factories, warehouses, manufacturing and production. A mention counts in every group whose words surround it, and in none if no business word is nearby. Table 2 shows one mention of each kind from the 10-Ks of well-known firms, namely Apple's sales in Greater China, Lowe's imports from China, Intuitive Surgical's plant in Mexico, and a risk-factor sentence that names Russia without any business word.

From these counts, the measure of a firm's sales exposure to the shocked country is

$$SalesExp_{is} = \ln\left(1 + 10{,}000 \times \frac{\text{mentions of the shocked country near sales words}}{\text{words in the 10-K}}\right), \quad (1)$$

for firm i and shock s. Sourcing exposure ($SourceExp_{is}$), operations exposure ($OpsExp_{is}$) and total exposure ($TotalExp_{is}$, all mentions) are defined in the same way. Scaling by length makes exposure comparable across 10-Ks of different sizes, and the logarithm limits the influence of the few firms with many mentions of a country. I call a firm sales-exposed when its 10-K in use mentions selling to the shocked country, as in 16 percent of the main sample's firm-shock observations (Table 3). I count mentions in the whole 10-K rather than in a section chosen in advance, and Section 6.4 compares the sections.

3.2 Analysts' forecasts

I use each analyst's own forecasts of annual street EPS from the I/B/E/S detail file (Bradshaw and Sloan, 2002). Forecasts and actuals are unadjusted for stock splits and checked for share-basis changes (Payne and Thomas, 2003; Robinson and Glushkov, 2006). The target year is the first fiscal year ending at least six months after the shock's peak month, so that

most of it falls after the shock.

For each analyst, I use four forecasts of the target year. The pre-shock forecast, F^0, is the analyst's last forecast before the shock's first month, issued or confirmed after the firm's last earnings announcement before the shock, so that it reflects the firm's latest reported earnings. The pre-EA forecast, F^{pre}, is the forecast in effect on the day before the firm's first earnings announcement (EA) after the shock's peak month, a median of 46 days after the end of the peak month. If the analyst has not revised since the shock began, it is the pre-shock forecast itself, and the revision is zero. The post-EA forecast, F^{post}, is the analyst's first forecast on or after that announcement, and the final forecast, F^{final}, is the analyst's last forecast before the target year's annual earnings announcement. Figure 2 places the four forecasts and the events around them on a timeline.

The three post-shock forecasts show when analysts incorporate the shock. The pre-EA forecast comes after every qualifying measure or war is public, which the selection requires by the end of the peak month (Section 2.1), and before the firm's own account of the shock, so it measures what analysts work out from public information. The post-EA forecast follows the firm's first earnings announcement after the shock's peak, when analysts can interpret what the firm reports (Chen et al., 2010). The final forecast shows how much they have incorporated by the end of the year.

All four forecasts come from the same analyst, and I drop forecasts that I/B/E/S excludes or that the broker has stopped. The pre-shock forecast error and the revisions are scaled by the share price before the shock and winsorized at the 1st and 99th percentiles within each shock, and each post-shock forecast error is their difference.

Some analyses use the consensus forecast instead, the mean estimate in the I/B/E/S monthly summary, at the firm level. For the pre-shock, pre-EA and final forecasts, the consensus is the last monthly mean before that point, and it is missing when that mean is more than 35 days old. For the post-EA forecast, it is the first monthly mean on or after the announcement. It differs from the mean of the analysts' own forecasts because I/B/E/S leaves out estimates that analysts have not updated or confirmed for 180 days (105 days before June 2010), whereas each analyst's own forecast is kept however old it is.

3.3 Sample and controls

The sample consists of U.S. firms with I/B/E/S coverage, linked to CRSP and Compustat with links valid on the day before the shock begins. The main sample keeps firm-shock observations of non-financial firms with a share price of at least $1, a pre-EA forecast within 120 days of the end of the peak month and no acquisition in the target year larger than 25

percent of the previous year's sales. Financial firms are those in industries 44 to 47 of the Fama–French 48 classification (Fama and French, 1997). The sample is then restricted to large and clean firms.

Large firms are those above the shock's median market value. The restriction focuses the test on well-covered firms, where analysts have the most resources and the strongest incentives to incorporate public information, and it keeps most of the market: the larger half holds about 98 percent of the market value of the firms ranked (Appendix A.4). Analysts are known to forecast small, thinly covered firms differently. They are more optimistic where information is harder to obtain (Lim, 2001), and news reaches the prices of small, little-covered stocks more slowly (Hong et al., 2000). The data bear this out. Among the smaller firms, which have a median of three analysts against eight among the larger ones, sales exposure predicts a lower pre-shock forecast error in each of the two years before the shock, not only in the shock year (Appendix A.4), whereas among the larger firms it predicts no shortfall in either earlier year (Table 5). Since slow incorporation is strongest among small firms (Zhang, 2006), leaving them out also works against finding under-reaction.

Clean firms have no restatement or non-reliance filing for the target or previous fiscal year (from Audit Analytics) and no special items larger than 5 percent of assets in the target year, since both distort the earnings that analysts forecast. The main sample has 29,177 firm-shock observations and 268,865 analyst-firm-shock observations across the 39 shocks. Appendix Table A2 traces the sample step by step, from every firm with a 10-K in use to the main sample.

The controls account for firm characteristics that predict forecast errors (So, 2013) and for the horizon of the forecast (Richardson et al., 2004). The six standard controls are the logarithm of market value, the number of countries the 10-K mentions, the logarithm of one plus mentions of all foreign countries per 10,000 words, the logarithm of the number of analysts, the months from the shock to the end of the target year, and the days from the end of the peak month to the pre-EA forecast. The six firm controls are return on assets, leverage, the book-to-market ratio, a loss indicator, five-year earnings volatility, and the stock return over the 12 months before the shock. The signed design includes both sets, and the unsigned design includes the six standard controls. Appendix Table A1 defines every variable, and Table 3 reports descriptive statistics.

4 Research Design

Within the same industry and the same shock, a firm's sales exposure to the shocked country predicts part of its pre-shock forecast error. The signed design estimates the share of that

predictable part that analysts' revisions incorporate, with an instrumental-variable (IV) estimate that uses sales exposure as the instrument. This section defines the incorporation rate formally, states the hypotheses of Section 2.3 in terms of the estimating equations, and explains how the rate is estimated and what it assumes. Section 6.3 checks the result with an unsigned design that does not assume the direction of the effect.

4.1 The incorporation rate

For analyst j, firm i in industry k and shock s, with actual EPS A_{is} for the target year, share price P_{is} before the shock, pre-shock forecast F^0_{ijs} and post-shock forecast F^r_{ijs} for $r \in \{\text{pre}, \text{post}, \text{final}\}$, define

$$FE^0_{ijs} = \frac{A_{is} - F^0_{ijs}}{P_{is}}, \qquad REV^r_{ijs} = \frac{F^r_{ijs} - F^0_{ijs}}{P_{is}}, \qquad FE^r_{ijs} = \frac{A_{is} - F^r_{ijs}}{P_{is}}. \tag{2}$$

These definitions split the pre-shock forecast error into the part the analyst revised away and the part that remains, so that $FE^0_{ijs} = REV^r_{ijs} + FE^r_{ijs}$.

The signed design rests on two regressions on sales exposure,

$$FE^0_{ijs} = \beta^{FE}\, SalesExp_{is} + \gamma_1 SourceExp_{is} + \gamma_2 OpsExp_{is} + \delta' X_{is} + \alpha_{ks} + \varepsilon_{ijs}, \tag{3}$$

$$REV^r_{ijs} = \beta^{REV}_r\, SalesExp_{is} + \lambda_{1r} SourceExp_{is} + \lambda_{2r} OpsExp_{is} + \kappa'_r X_{is} + \tilde{\alpha}^r_{ks} + v^r_{ijs}. \tag{4}$$

Here $SalesExp_{is}$, $SourceExp_{is}$ and $OpsExp_{is}$ are firm i's sales, sourcing and operations exposure to the country of shock s, counted in its 10-K in use (Section 3.1), and α_{ks} and $\tilde{\alpha}^r_{ks}$ are industry-by-shock fixed effects, with Fama–French 48 industries. The controls X_{is} are the twelve of Section 3.3: the firm's size and analyst coverage, how much its 10-K says about foreign countries, the horizon and timing of the forecasts, and six accounting and market characteristics that predict forecast errors.

Both regressions hold sourcing and operations exposure fixed because firms' ties overlap. Firms that sell in a country often operate there too, and the correlation between sales and operations exposure is 0.75. Operations exposure can push earnings the other way (Section 2.2), and Section 5.1 shows what happens to β^{FE} when it is left out. Like sales exposure, both are measured before the shock, so neither is an outcome of it.

In these terms, H1 predicts that sales exposure lowers the pre-shock forecast error, $\beta^{FE} < 0$, and H2 that it lowers the revision, $\beta^{REV}_r < 0$ (Section 2.3). The incorporation rate compares the two slopes. If a unit of sales exposure changes the pre-shock forecast error by β^{FE} and the revision by β^{REV}_r, analysts have revised away the share $\pi_r = \beta^{REV}_r / \beta^{FE}$ of the exposure-driven forecast error by forecast r. Because $FE^0_{ijs} = REV^r_{ijs} + FE^r_{ijs}$, the same

regression for FE^r has coefficient $\beta^{FE} - \beta_r^{REV}$ on sales exposure, the part that remains. H3, the efficiency benchmark, is $\pi_r = 1$, under which that coefficient is zero, and under-reaction gives $0 < \pi_r < 1$.

The ratio is an instrumental-variable estimate. It equals the coefficient π_r in

$$REV^r_{ijs} = \pi_r \, FE^0_{ijs} + \mu_{1r} SourceExp_{is} + \mu_{2r} OpsExp_{is} + \psi_r' X_{is} + \bar{\alpha}^r_{ks} + u^r_{ijs}, \tag{5}$$

estimated by two-stage least squares (2SLS) with sales exposure as the excluded instrument for FE^0_{ijs}, so that equations (3) and (4) are its first stage and reduced form. Writing the rate this way gives it the standard errors and weak-instrument-robust inference of IV (Section 4.2).

Why instrument the forecast error at all? An OLS regression of REV^r on FE^0 measures how revisions track forecast errors in general, and most of the pre-shock forecast error comes from news that arrives after the revision, which no revision can contain. Noise in the pre-shock forecast itself also enters both variables, with the same sign. The instrument avoids both problems, because it uses only the part of the forecast error that the shock and the firm's 10-K predict, and both were public when the shock peaked. Table 4 reports the OLS coefficient alongside the IV estimate.

The IV estimate is therefore an incorporation rate, not a causal effect of forecast errors on revisions. Unlike the bias or accuracy of the forecasts after the shock, it scales the revision by the size of the shock's predictable effect on earnings, so it can be compared across forecasts and with other settings. Its complement, $1 - \pi_r$, is the under-reaction, and the ratio $(1 - \pi_r)/\pi_r$ is the coefficient from regressing the forecast error FE^r on the revision (Coibion and Gorodnichenko, 2015), restricted to variation driven by sales exposure.

4.2 Estimation and identification

Because the 39 shocks are stacked and the fixed effects are by industry and shock, each coefficient is a weighted average of comparisons between firms that face the same shock in the same industry but differ in how much they sell to the shocked country (Gormley and Matsa, 2011). Exposure is measured from a 10-K filed before the shock, the shocks are foreign events that U.S. firms do not choose, and the fixed effects absorb anything common to an industry within a shock, including industry-wide changes in demand or costs. The regressions weight observations so that each shock counts equally, and each firm equally within its shock, so analysts covering the same firm share that firm's weight. Because outcomes are correlated across firms that face the same event (Bernard, 1987), and forecast errors are no exception (Keane and Runkle, 1998), I cluster standard errors by shock. With 39 clusters, the t critical values use 38 degrees of freedom (Cameron and Miller, 2015).

The IV estimate rests on three conditions. The first is relevance, which requires that sales exposure predict the pre-shock forecast error once sourcing and operations exposure are included. The strength of the instrument also matters for inference. With one instrument and clustered errors, the effective first-stage F-statistic of Montiel Olea and Pflueger (2013) equals the squared clustered t-statistic of the first stage. In the main specification it is about 12 (Section 5), above the usual threshold of 10 (Staiger and Stock, 1997) but too low for ordinary t-based intervals to have their nominal coverage (Lee et al., 2022). I therefore report 95% Anderson–Rubin sets, which remain valid however weak the instrument is (Anderson and Rubin, 1949; Andrews et al., 2019). Throughout, brackets after an incorporation rate give this set, and brackets after a difference between two rates give its 95 percent confidence interval.

The second is the exclusion restriction, which here requires that sales exposure be related to the pre-shock forecast error and the revision only through what the shock does to the firm's target-year earnings. Testing whether π_r equals one does not need it. Under efficient forecasts no public information, sales exposure included, predicts the error that remains after the revision (Section 2.3), so sales exposure is then a valid instrument by construction. The restriction matters for the size of π_r.

It could fail in two ways, and only the first could create under-reaction. Sales exposure could be related to the error the forecast would have had anyway, from noise in the forecast or news after the revision, so that sales-exposed firms would have fallen short even without the shock. Section 6 tests this with the same firms in the years before the shock and with exposure to countries that were not shocked. Later developments of the shock, such as an escalation, do not violate the restriction when they could be expected, because efficient forecasts would already include them. Those no one could expect average out across shocks, which is why inference is by shock (Section 6.5). The second way is that analysts cut their forecasts for sales-exposed firms over risks that never reach target-year earnings. That would deepen the revision without enlarging the shortfall and so raise π_r, understating the under-reaction rather than creating it. Measurement error in sales exposure, as long as it is unrelated to earnings news, shrinks both slopes by the same factor and leaves π_r unchanged.

The third is monotonicity, which here means that sales exposure lowers the pre-shock forecast error in every shock, not only on average. If its effect differs across shocks, π_r averages shock-level rates with weights proportional to each shock's first stage, and the weights are all nonnegative only if this holds. Section 6.5 reports the estimate without the shocks whose weights are negative, and Appendix A.4 the weights themselves.

5 Main Results

Sales-exposed firms fall short of their pre-shock forecasts, and analysts revise those forecasts downward. By the pre-EA forecast, the last before the firm's first earnings announcement after the shock, the revisions incorporate about a third of the exposure-driven forecast error, and by the final forecast, the last before the annual announcement, about two-thirds. This section reports these estimates, translates them into earnings per share, and follows the incorporation rate from the start of the shock to the end of the year.

5.1 The incorporation rate before the firm reports

The more a firm sells to the shocked country, the further its earnings fall short of analysts' pre-shock forecasts. In the main specification, column (3) of Table 4, Panel A, each unit of sales exposure lowers the pre-shock forecast error by 2.42 thousandths of the share price (the first stage, equation (3); $t = -3.49$, effective first-stage $F = 12.2$), as H1 predicts. Analysts do respond. By the pre-EA forecast, they have cut their forecasts for these firms by 0.87 thousandths of the share price per unit of exposure (the reduced form, equation (4); $t = -3.67$), as H2 predicts. The cut covers only part of the shortfall, however, and the rest is still there in the pre-EA forecast error, which also falls with sales exposure (-1.55, $t = -2.39$).

Dividing the cut by the shortfall, the reduced form by the first stage, gives the incorporation rate. At the pre-EA forecast it is $\pi_{\text{pre}} = (-0.868)/(-2.419) = 0.36$, with a 95% Anderson–Rubin set of $[0.17, 0.79]$. Before the firm next reports, analysts' revisions thus incorporate about a third of the exposure-driven forecast error. The set excludes one, the value under efficient forecasts, and zero, the value if analysts ignored the shock. Analysts revise the forecasts of the right firms in the right direction, but by too little, and the under-reaction before the firm reports, $1 - \pi_{\text{pre}}$, is 0.64.

The under-reaction is similar in size to that found in professional forecasters' inflation forecasts. A positive coefficient from regressing the forecast error on the revision is the signature of sluggish updating, both in professional forecasts (Coibion and Gorodnichenko, 2015) and in analysts' firm-level forecasts (Bouchaud et al., 2019). Here that coefficient, $(1-\pi_{\text{pre}})/\pi_{\text{pre}}$, is the pre-EA forecast error's slope on sales exposure divided by the revision's, $1.55/0.87 \approx 1.8$, so for each unit of exposure-driven revision almost two units of the forecast error remain. That is close to the 1.91 to 2.20 that Coibion and Gorodnichenko (2015) find for professional forecasters' GDP-deflator inflation forecasts when they instrument the revisions with oil-price changes, the closest counterpart to the IV estimate here, and above the 1.19 of their OLS estimate. Unlike those estimates, it concerns a specific, public foreign

event and only the consequences for each firm that its own 10-K makes predictable.

Without the instrument, analysts would appear to incorporate even less. The OLS coefficient of the pre-EA revision on the pre-shock forecast error is 0.18 ($t = 9.72$ in Table 4, Panel B), half the IV estimate. Most of the pre-shock forecast error comes from news that arrives after the revision (Section 4.1), which no revision could contain, so OLS understates how much of the predictable error analysts incorporate.

As Section 4.1 anticipated, holding operations exposure fixed matters. In column (3), operations exposure predicts a higher pre-shock forecast error (3.39, $t = 3.64$) and an upward revision (0.99, $t = 2.26$), while sourcing exposure has no detectable effect (-1.64, $t = -0.56$). Leaving operations exposure out therefore lets the gains of firms that operate in the shocked country offset the losses of firms that sell there, as the channels of Section 2.2 imply. Total exposure, in column (1), has no first stage at all (0.18, $t = 0.38$), and sales exposure alone, in column (2), has a first stage of the predicted sign but a weak one (-0.46, $t = -0.86$, effective first-stage $F = 0.7$). Holding operations exposure fixed sharpens both stages while leaving their ratio similar, as the IV estimate is 0.44 in column (2) and 0.36 in column (3).

5.2 The shortfall in earnings

Translated into earnings per share, the exposure-driven shortfall of the median sales-exposed firm is about 1.7 percent of annual EPS. Sales-exposed firms are 16 percent of the sample (Table 3), and the median sales-exposed firm has sales exposure of 0.39, about one mention of selling to the shocked country per 20,000 words. Its pre-shock forecast error is lower than that of a firm in the same industry without sales exposure by $2.42 \times 0.39 \approx 0.95$ thousandths of the share price. That is about 5.5 cents a share at the median share price of \$58, or 1.7 percent of annual EPS at the median earnings yield of 5.6 percent. At the 90th percentile of sales-exposed firms, with sales exposure of 1.23, the shortfall is 3.0 thousandths of the share price, about 5 percent of annual EPS. For the median sales-exposed firm, analysts' revisions incorporate 0.34 thousandths of the share price by the pre-EA forecast, and the other 0.61 thousandths remain in the pre-EA forecast error.

These shortfalls are small next to the typical absolute pre-shock forecast error for sales-exposed firms (a median of 6.0 thousandths of the share price), most of which no public information predicts. What matters, though, is that the exposure-driven part was predictable from the shock and the firm's 10-K, both public when the shock peaked, yet analysts incorporated only about a third of it before the firm reported.

5.3 How quickly analysts incorporate the shock

Analysts incorporate the shock gradually. Figure 3, Panel B, follows the incorporation rate from the start of the shock to the pre-EA forecast, using each analyst's forecast in effect on each day. Because the first stage is the same on every day, the path traces the revisions alone. For the first two weeks, analysts incorporate nothing (-0.01 at day 15). After a month they have incorporated 0.08, after six weeks 0.20 and after three months 0.26, and by the pre-EA forecast the rate reaches 0.36.

Over the rest of the year, forecasts catch up. The consensus forecast (Section 3.2) goes from incorporating 0.38 before the firm reports to 0.92 $[0.81, 1.11]$ by the end of the year (Table 5). Analysts' own forecasts go from 0.36 at the pre-EA forecast to 0.44 $[0.17, 0.99]$ at the post-EA forecast and 0.69 $[0.43, 0.93]$ at the final forecast (Table 4, Panel B, and Figure 4). Most of this catching up comes after the firm's first earnings announcement rather than with it. Of the rise from 0.36 to 0.69, about a quarter comes with the announcement and its conference call, where analysts can ask management about the shock, and the other three-quarters over the rest of the year.

Even so, analysts' own final forecasts fall short. Their Anderson–Rubin set excludes one, and the final forecast error still falls with sales exposure (-0.76, $t = -2.38$). The consensus does better because it leaves out forecasts that analysts have stopped updating (Section 3.2). Analysts who issued or confirmed their final forecast in the 120 days before the annual announcement incorporate 0.86 $[0.68, 1.18]$, close to the consensus (Appendix Table A12, Panel C).

6 Robustness

Section 5 compares firms that face the same shock in the same industry but differ in their sales exposure to the shocked country. Five alternatives could produce its estimates instead of the shock itself. Analysts could be routinely too optimistic about sales-exposed firms, shock or no shock (Section 6.1), or the estimates could reflect a general feature of firms that sell abroad rather than anything about the shocked country (Section 6.2). Either would mean that sales-exposed firms fall short of their forecasts even without the shock, the failure of the exclusion restriction that Section 4.2 singles out. The estimates could also be an artifact of the signed design, which assumes the direction of the effect (Section 6.3), come from a text measure that picks up something other than where firms sell (Section 6.4), or be due to chance or to a few shocks, since inference rests on 39 clusters (Section 6.5). None of the five accounts for the estimates.

6.1 Years without a shock and the months before it

If analysts were routinely too optimistic about sales-exposed firms, sales exposure would predict lower forecast errors in every year, not only after a shock. Table 5 checks this by repeating the analysis at the firm level, with the consensus forecast, for the same firms in the shock year and in each of the two years before it, taking forecasts at the same points in the calendar shifted back by one and two years. In those earlier years sales exposure should predict neither forecast errors nor revisions, although such pre-period tests have limited power (Roth, 2022).

Both the shortfall and the revision are specific to the shock year. In that year, sales exposure predicts a lower pre-shock forecast error (-3.35, $t = -4.18$) and a downward revision (-1.26, $t = -4.67$), an incorporation rate of 0.38 $[0.20, 0.76]$ that is close to the analyst-level estimate. In the year before the shock, neither slope is significant (-0.05, $t = -0.04$, and 0.29, $t = 0.67$), and two years before, both are smaller and not significant (-1.51, $t = -1.39$, and -0.85, $t = -1.54$). Measured against the mean of the two earlier years, the shock-year slopes are -2.57 ($t = -2.45$) and -0.98 ($t = -2.53$), with an incorporation rate of 0.38 $[0.08, 2.12]$. Among the large firms of this sample, there is thus no standing optimism about sales-exposed firms of the kind that appears among smaller firms (Section 3.3), and analysts do not routinely revise their forecasts for sales-exposed firms downward.

Nor did analysts revise significantly before the shock. Had they anticipated it, the pre-shock forecast would already contain part of the shortfall, and π_{pre} would understate how much they incorporate. From each analyst's forecast 180 days before the shock's first month to the pre-shock forecast, revisions do not vary with sales exposure (slope -0.43, $t = -0.75$), and no month before the shock has a significant slope (Appendix Figure A3). After the shock begins, the exposure-driven revision first differs from zero at the 5 percent level on day 45, and it reaches the -0.87 of Table 4 by the pre-EA forecast (Figure 3, Panel A). Measured against the forecast 180 days before the shock, the part of the exposure-driven shortfall already in the pre-shock forecast is 0.14 $[-0.40, 0.45]$. On the 77 percent of forecasts that reach back that far, π_{pre} is 0.37 (0.36 in the full sample), and counting this small and insignificant pre-shock drift as incorporation would raise it only to 0.45 $[0.03, 0.83]$, a set that still excludes one.

6.2 Placebo countries

If the estimates reflected a general feature of firms that sell abroad, exposure to countries that were not shocked should produce similar results. It does not. The placebo countries are the 20 non-U.S. countries named in the most U.S. 10-Ks, counted over every 10-K in

the sample period, in the spirit of the placebo tests of Bertrand et al. (2004). I replace each firm's sales, sourcing and operations exposure to the shocked country with its exposure to each of them, leaving out any shocks to the placebo country itself. A placebo country is as strong as the shocked country when its t-statistic is at least as large in absolute value, and as strong in the same direction when it also has the same sign.

No placebo country is as strong as the shocked country in the same direction on the pre-shock forecast error, the pre-EA revision or the pre-EA forecast error (Table 6, Panel B, with every country listed in Appendix Table A7). The closest are China, whose exposure predicts the pre-shock forecast error with a t-statistic of -2.37, and Canada and Israel, whose exposure predicts the revision with -1.78, against -3.49 and -3.67 for the shocked country. Only countries with the opposite sign are as strong, chiefly the Netherlands, whose exposure predicts higher forecast errors with no effect on the revision, and no country matches the shocked country on all three outcomes, in either direction.

6.3 A design that does not assume the direction

The signed design relies on the direction in which sales exposure moves earnings. A design that drops this assumption gives a similar answer. The unsigned design works with the size of forecast errors instead, since a shock can create winners and losers among exposed firms (Lang and Stulz, 1992; Guidolin and La Ferrara, 2007) and errors in either direction increase absolute forecast errors.

For firm i and shock s, the unsigned design uses I/B/E/S consensus forecasts $\bar{F}$ in the shock year ($t = 0$) and, for the same firm, at the same points in time one and two years earlier ($t = -1, -2$). Each year gives the absolute pre-shock forecast error $AFE^0_{ist} = |A_{ist} - \bar{F}^0_{ist}|/P$, its counterpart AFE^r_{ist} at forecast r, and the accuracy improvement $IMP^r_{ist} = AFE^0_{ist} - AFE^r_{ist}$, the reduction in the absolute forecast error that the revision achieves. Because sales-exposed firms' absolute forecast errors can differ from other firms' even in years without a shock, each is measured against the firm's own normal level,

$$\Delta AFE_{is} = AFE^0_{is0} - \tfrac{1}{2}\left(AFE^0_{is,-1} + AFE^0_{is,-2}\right), \tag{6}$$

and likewise ΔIMP^r_{is}. The unsigned incorporation rate $\theta_r = \beta^{\Delta IMP}_r / \beta^{\Delta AFE}$ is the IV estimate from regressing ΔIMP^r_{is} on ΔAFE_{is}, with sales exposure as the excluded instrument and the six standard controls and industry-by-shock fixed effects in both stages. Sales exposure enters alone, because in absolute errors a loss and a gain both increase the error, so the offsetting channels that lead the signed design to include sourcing and operations exposure do not cancel. The sample keeps large firms without a restatement or non-reliance filing, in

all industries. The design uses the consensus because it gives a forecast at the same point in every year, and the choice does not matter, since with the consensus the signed design gives nearly the same rate as with analysts' own forecasts (Section 6.1).

In the main specification (Table 7, column (2)), sales exposure enlarges the absolute pre-shock forecast error relative to the firm's two earlier years (a first stage of 1.93, $t = 3.10$, effective first-stage $F = 9.6$) and increases the accuracy improvement by the pre-EA forecast (a reduced form of 0.59, $t = 2.43$). The IV estimate is $\theta_{\text{pre}} = 0.31$ $[0.05, 0.93]$, against an OLS coefficient of 0.20. Because it is measured in absolute errors, θ_r is not the same quantity as π_r. Before the firm reports, it can only overstate how much analysts incorporate (Appendix A.4), so its set, which excludes one, is conservative evidence of under-reaction. Like the signed rate, the unsigned rate rises over the year, to 0.52 $[0.21, 1.36]$ at the post-EA forecast and 0.91 $[0.59, 1.07]$ at the final forecast (Panel B and Figure 4). No placebo country is as strong as the shocked country on both outcomes (Table 6), and Appendix A.4 reports the design's other baselines.

The unsigned design also explains why total exposure fails in the signed design. Total exposure, which has no first stage there, enlarges absolute forecast errors here (1.38, $t = 3.36$), because in absolute value the losses of firms that sell in the shocked country and the gains of firms that operate there add up rather than cancel.

6.4 Exposure by 10-K section and other mentions

The text measure picks up what firms say about selling in the country, not merely that they name it. Counted one section of the 10-K at a time, sales exposure predicts most strongly in the MD&A and not at all in the risk factors, the pattern expected of a measure of where firms sell. The main measure counts mentions anywhere in the 10-K, and Table 8 counts them separately in each section, for the firms whose 10-K has that section. In the MD&A (Item 7), where SEC rules require managers to discuss known trends and uncertainties that are reasonably likely to affect sales and income (U.S. Securities and Exchange Commission, 2021), the signed design's slopes are -1.60 ($t = -2.99$) for the pre-shock forecast error and -0.48 ($t = -3.27$) for the revision, with $\pi_{\text{pre}} = 0.30$ $[0.12, 0.84]$, and no placebo country is as strong in the same direction on the forecast error. In the business description (Item 1), the signed slopes are weaker ($t = -1.64$ and -2.28), the first stage is weak (effective first-stage $F = 2.7$), and several placebo countries are as strong. No single section is as strong as the whole 10-K ($t = -3.49$ and -3.67), which counts the mentions in all of them.

In the risk factors (Item 1A), the signed slopes have the wrong sign ($t = 1.53$ and 1.03). The risk factors list what could make an investment risky (U.S. Securities and Exchange

Commission, 2020) and have grown longer and more standardized over time (Dyer et al., 2017). A country named among them tells the reader that the firm could be affected, not that it sells there. In the unsigned design, too, the risk factors predict nothing ($t = 0.41$), while the MD&A and Item 1 enlarge absolute forecast errors about equally ($t = 2.16$ and 2.19).

A last check holds the firm and the country fixed and changes only what the 10-K says about the country. Mentions of the shocked country with no sales, sourcing or operations words nearby predict nothing in the signed design (Table 8, column (5)). In place of sales exposure, their first-stage slope has $t = -1.09$ (effective first-stage $F = 1.2$), and added to the main specification they leave π_{pre} at 0.34 $[0.07, 0.94]$. In the unsigned design, they enlarge absolute forecast errors as much as sales exposure does ($t = 3.39$ against 3.10), but they predict nothing among firms that do not sell there, so their effect comes from sales-exposed firms. The unsigned design therefore cannot separate the sales words from the depth of a sales-exposed firm's ties to the country, whereas the signed design can.

6.5 Inference and influential shocks

Chance does not explain the estimates either. With only 39 clusters, clustered standard errors can overstate precision (Cameron and Miller, 2015), but the wild cluster bootstrap and the studentized randomization test in Table 6, Panel A, give much the same p-values. For the signed design's slopes of the pre-shock forecast error, the pre-EA revision and the pre-EA forecast error, the clustered p-values are 0.001, 0.001 and 0.022. The restricted wild cluster bootstrap, with the six-point weights of Webb (2023), gives 0.001, 0.003 and 0.020 (Cameron et al., 2008). Randomization inference, which shuffles exposure across firms within industry-by-shock cells, gives 0.002, 0.002 and 0.016 for the studentized statistic (Young, 2019), and 0.025, 0.022 and 0.097 for the slope itself. In the unsigned design, the wild cluster bootstrap gives 0.002 for the first stage and 0.024 for the reduced form.

No single shock drives the incorporation rate (Appendix Table A8). Leaving out one shock at a time, π_{pre} ranges from 0.30, without Canada in 2018, to 0.44, without Mexico in 2009, and every Anderson–Rubin set excludes zero. Leaving out groups of shocks that overlap in time gives 0.25 to 0.44, and clustering by the 19 overlap groups instead of the 39 shocks gives t-statistics of -3.22 for the pre-shock forecast error and -3.64 for the revision, with a set of $[0.16, 0.93]$. In the unsigned design, leaving out one shock at a time keeps θ_{pre} between 0.25 and 0.41.

The estimate also does not depend on shocks in which the instrument works the other way. Sales exposure lowers the pre-shock forecast error in 26 of the 39 shocks. Written as a

weighted average of shock-level rates, with weights proportional to each shock's first stage, π_{pre} gives the other 13 small negative weights, -0.25 in total, and without them the rate is 0.32 (Appendix A.4). The weight is concentrated, with an effective number of about eight shocks, which is why the leave-one-out estimates matter. Section 7.5 estimates the 24 shocks without war on their own and compares them with the 15 that involve war.

7 Why Do Analysts Under-react?

Before the firm reports, analysts incorporate about a third of the exposure-driven forecast error (Section 5). This section asks why. It takes four explanations in turn: inattention (Section 7.1), exposure that is hard to find (Section 7.2), a reluctance to deliver bad news (Section 7.3) and a lack of experience with such shocks (Section 7.4). Each explanation names a group in which the under-reaction should be much smaller, namely analysts who follow the shock, firms whose exposure is easy to see, good news, and analysts with experience of an earlier episode in the same country. Each test compares that group with the rest. In none of the four is the under-reaction much smaller, so none of the explanations accounts for it. What remains, described in Section 7.5, is a shared under-adjustment: analysts revise as if the shock's expected effect on earnings were smaller than it is. They revise in the right direction but too little, their forecasts do not spread apart, and they revise less after wars, whose consequences are hardest to gauge, than after economic measures.

7.1 Noticing the shock

Analysts notice the shock, so inattention does not explain the under-reaction. If analysts did not notice the shock, or did not get around to revising their forecasts (Sims, 2003; Hirshleifer et al., 2009; Driskill et al., 2020), the under-reaction would come from forecasts left unrevised. The analysts who have revised would then incorporate most of the exposure-driven forecast error, and analysts who pay attention to the shock would incorporate more of it than their colleagues.

Analysts revise their forecasts for sales-exposed firms sooner than those for other firms (Figure 3, Panel C). By day 60, 65 percent have revised, against 57 percent. They notice that these firms are affected, but they revise too little, since over the same weeks the incorporation rate rises only gradually, to 0.36 by the pre-EA forecast (Section 5.3).

Nor do forecasts that analysts have not yet revised explain much of the under-reaction before the firm reports, unlike at the end of the year (Section 5.3). By the pre-EA forecast, 21 percent of analyst-firm-shock observations have not been revised (26 percent weighted

as in Figure 3, Panel C). The analysts who have revised incorporate 0.48 $[0.20, 2.22]$ of the exposure-driven forecast error, against 0.36 for all analysts. The difference is not significant ($t = 0.96$).

A direct measure of attention gives the same picture. It marks analysts who revised the forecast of another sales-exposed firm in the first 30 days of the shock, outside that firm's own earnings announcement, and so were following the shock. Compared with colleagues covering the same firm in the same shock, these analysts are 4.4 percentage points more likely to have revised by the pre-EA forecast ($t = 7.82$), but they incorporate no more of the exposure-driven forecast error on any day. By the pre-EA forecast, the difference is -0.10, with a 95 percent confidence interval of $[-0.31, 0.11]$. Attention changes when analysts revise, not how much of the shock's effect on earnings they incorporate.

7.2 Finding the exposure

Analysts notice the shock and know which firms it affects, but they could still underrate how much a firm sells in the shocked country if that exposure is hard to find. That is not the reason for the under-reaction either. Mentions of a country spread through a 10-K may be easy to overlook, whereas a figure for the country's sales is not. If analysts were slow because they missed the exposure, they would incorporate more when it is visible. For 551 of the 4,811 sales-exposed firm-shocks, in 29 shocks, the firm reports a geographic segment named for the shocked country in the fiscal year of its 10-K in use (Compustat historical segments, as first reported). I estimate the signed design twice, once for these firms and once for the other sales-exposed firms, each time together with the firms that do not sell in the shocked country.

Visible exposure is not incorporated faster. By the pre-EA forecast, firms with visible exposure have a rate of 0.21 $[-0.14, 0.42]$, against 0.37 $[0.12, 1.17]$ for the other sales-exposed firms, a difference of -0.16 $[-0.47, 0.14]$ (Appendix Table A12, Panel A). After the firm reports the two are close, at 0.41 and 0.44 by the post-EA forecast and 0.61 and 0.68 by the final forecast.

7.3 Good news and bad news

Analysts who can find the exposure might still hold back from cutting their forecasts. They are no faster with good news than with bad, however, so a reluctance to deliver bad news does not explain the under-reaction. Reluctance is a long-standing explanation for analysts' slow response to bad news. Analysts' forecasts tend to be optimistic, because brokerages reward optimism (Hong and Kubik, 2003) and cutting a forecast can strain an analyst's relationship

with management (Lim, 2001). Easterwood and Nutt (1999) find that analysts under-react to negative information and over-react to positive information. If reluctance explained the under-reaction here, analysts would incorporate good news from the same shocks faster than bad news, and might even over-react to it.

The same shocks also bring good news, which makes a direct test possible. In the main specification, operations exposure raises the pre-shock forecast error (3.39, $t = 3.64$, Table 4). Firms that operate in the shocked country thus earn more than analysts expected, consistent with lower local costs (Section 2.2). Using operations exposure as the instrument in place of sales exposure, with sales and sourcing exposure in both stages, gives the incorporation rate of this good news (Table 9). Before the firm reports, analysts incorporate 0.29 $[0.05, 0.50]$ of it, against 0.36 $[0.17, 0.79]$ of the bad news from sales exposure, and by the final forecast they incorporate 0.78 $[0.55, 1.01]$, against 0.69 $[0.43, 0.93]$. At none of the three forecasts can the two rates be told apart (p-values of 0.556, 0.639 and 0.236), and with both exposures as instruments, the common rate before the firm reports is 0.30 $[0.13, 0.59]$. Nor do analysts over-react to the good news. They leave part of it in their forecast errors, as they do the bad news. The pre-EA forecast error rises with operations exposure ($t = 3.44$), and a rate of one is rejected ($p = 0.001$).

The operations result passes most of the checks of Section 6. No placebo country is as strong in the same direction on the pre-shock or the pre-EA forecast error, although three are on the revision. Against the years without a shock it is weaker, because operations exposure leans toward higher forecast errors in the two earlier years as well, and the shock-year slope measured against them has $t = 1.83$.

There is thus no sign that the under-reaction reflects a reluctance to cut forecasts. The symmetry between good and bad news differs from analysts' response to aggregate macroeconomic news, where their forecast errors follow bad news but not good (Hugon et al., 2016). After a geopolitical shock, analysts get wrong how far earnings will move, not in which direction.

7.4 Experience

If analysts get the direction right and the size wrong, they may not yet have learned how far such a shock moves earnings, which experience could teach. Experience of an earlier episode in the same country does not make analysts faster before the firm reports, however, so the under-reaction does not come from a lack of experience either. Analysts' forecasts improve with experience (Mikhail et al., 1997; Clement, 1999), experienced analysts under-react less to past earnings (Mikhail et al., 2003), and analysts who covered firms exposed to earlier

epidemics forecast more accurately during the COVID-19 pandemic (Agarwal and Chalwati, 2023). Experience of past events also shapes how people form expectations (Malmendier and Nagel, 2011, 2016), and predictable forecast errors can arise while forecasters are still learning about an unfamiliar relation (Markov and Tamayo, 2006). If analysts learn from each episode how such shocks reach earnings, those who have seen one in the same country should incorporate more of the next.

An analyst has same-country experience if, during an earlier, finished episode of economic measures or war that directly involved the same country (one of the 102 such episodes in Appendix A.2), the analyst forecast a firm whose 10-K mentioned selling there, and that firm's annual earnings were announced before the new shock began (Appendix Table A1). Table 10 compares analysts with and without this experience in two ways that hold the rest fixed. Panel A compares colleagues covering the same firm in the same shock, and Panel B compares the same analyst across shocks, setting shocks to countries the analyst has experienced against the analyst's other shocks.

Neither comparison finds experienced analysts faster before the firm reports. Experienced analysts incorporate about as much as colleagues covering the same firm, with a difference of -0.01 $[-0.21, 0.19]$ by the pre-EA forecast and 0.03 $[-0.25, 0.31]$ by the final forecast. Two further comparisons in Panel A give the same answer. How much experience an analyst has does not matter, since one standard deviation more experience of earlier episodes, in any country, changes the rate by 0.00 $[-0.12, 0.11]$ before the firm reports. Nor do analysts at the largest brokerages incorporate more than their colleagues, although these brokerages are the most likely to employ economists and strategists who could explain how such shocks reach earnings. The difference is -0.03 $[-0.17, 0.12]$ before the firm reports and -0.02 $[-0.24, 0.21]$ at the final forecast.

Comparing the same analyst across shocks, the difference before the firm reports is 0.16 $[-0.09, 0.41]$, which is not significant. Experience may help later in the year. By the final forecast, the same analyst incorporates 0.36 $[0.01, 0.71]$ more of the error in a shock to a country the analyst has experienced, and the final forecast error is smaller ($t = 2.05$).[2]

[2]In the checks of Section 6, when experience of a placebo country replaces same-country experience, 2 of 16 placebo countries give an estimate as strong in the same direction, and when exposure to a placebo country replaces exposure to the shocked country, none of 20 does. Four of the 20 placebo countries cannot be used for experience: Switzerland, Spain and Singapore have no earlier episode, and experience of the Netherlands never varies within an analyst. Without each shock in turn, the estimate stays between 0.26 and 0.44 and is significant in 18 of 39 cases. Shuffling experience across analysts who cover the same firm in the same shock gives $p = 0.072$.

7.5 What remains: a shared under-adjustment

Analysts notice the shock, can find the firms' exposure and revise the forecasts of the right firms in the right direction, for good news as for bad, yet before the firms report they incorporate about a third of the exposure-driven forecast error, whatever their attention or experience. Because an analyst who is unsure of a shock's effect should still move the forecast by its expected value (Section 2.3), the evidence points to an expected effect that is too small. Two features of the evidence show that analysts adjust together and that they revise less when the effect is harder to judge.

The first is that analysts' forecasts do not spread apart. The under-reaction is not an average of analysts who incorporate the shock and others who ignore it: the dispersion of forecasts for sales-exposed firms does not widen after the shock, as it would if some analysts incorporated much more of it than others. Measured as the I/B/E/S consensus standard deviation over the share price, for the firms of Table 5 with at least three estimates, the change in dispersion from before the shock to the pre-EA forecast does not rise with sales exposure (a slope of -0.04 thousandths of the share price per unit of exposure, $t = -0.35$, with a 95 percent confidence interval of $[-0.25, 0.18]$, against a shortfall of 3.35 per unit). It does not rise by the post-EA or the final forecast either ($t = 0.73$ and 0.26), and the standard deviation across each firm's own analysts gives the same answer (Appendix Table A12, Panel B). Analysts adjust together, and by too little.

The second is the type of shock. After wars, whose economic consequences are harder to gauge than those of sanctions or tariffs, analysts revise less. Split by type (Appendix Table A9), the 24 shocks with economic measures and no war give $\pi_{\text{pre}} = 0.47$ $[0.25, 1.02]$ and $\pi_{\text{final}} = 0.81$ $[0.61, 1.09]$. For the 15 war shocks, sales exposure predicts a shortfall of similar size (a first-stage slope of -2.60 against -2.33), although too imprecisely (effective first-stage $F = 1.6$) to estimate the rate, and analysts' revisions respond less than after economic measures (a difference of 1.59 in the revision slope, $t = 3.16$). Analysts under-react more where information is more uncertain (Zhang, 2006), and geopolitical shocks give a case in which the harder the consequences are to gauge, the less analysts move.

This is the sluggish updating of Section 5.1, in a setting where the news is public, its direction is clear and its consequences for each firm are predictable from the firm's own 10-K. Forecasts then catch up over the rest of the year (Section 5.3), but before the firm reports, the forecasts that investors see stay predictably too high for the firms most exposed.

8 Conclusion

This paper asks whether sell-side analysts understand what geopolitical shocks do to the earnings of the firms they cover, and it measures that understanding by how much of a shock's predictable effect on earnings their forecast revisions incorporate. Across 39 shocks of economic measures or war to 22 countries, that effect comes from where firms sell. Within the same industry and shock, the greater a U.S. firm's sales exposure to the shocked country, measured from the 10-K it filed before the shock, the further its earnings fall below analysts' pre-shock forecasts. The shortfall is specific to the shock, because the same firms show no comparable pattern in the years before it and exposure to other countries produces nothing as strong in the same direction. Analysts revise toward the shortfall, but by too little. Before the firm reports, their revisions incorporate about a third of this shortfall, and by the final forecast about two-thirds. Analysts understand which firms a geopolitical shock hurts, but they underestimate how much it hurts them.

The pattern of the evidence also shows where the difficulty lies. Each explanation the paper considers names a group in which the under-reaction should be much smaller, and in none is it. Analysts revise their forecasts for sales-exposed firms sooner than for other firms, and those who follow the shock incorporate no more than colleagues covering the same firm, so the shock does not go unnoticed. Analysts incorporate no more when a firm reports the shocked country as a segment of its own, so the exposure is not hard to find. They incorporate good news from the same shocks no faster than bad, so the under-reaction shows no sign of a reluctance to deliver bad news. And neither experience of an earlier episode in the same country nor working at one of the largest brokerages, which are the most likely to employ economists, raises how much analysts incorporate before the firm reports. What analysts get wrong is not whether the shock matters or where the firm sells, but how far a shock to that country lowers the firm's earnings, a translation that falls to the analyst who covers the firm.

What remains is a shared under-adjustment: analysts revise as if the shock's expected effect on earnings were smaller than it is. They move their forecasts in the right direction but too little, they move together rather than spreading apart, and they revise less after wars, whose consequences are hardest to gauge, than after economic measures. Forecasts catch up over the rest of the year, but in the meantime the forecasts that investors see stay predictably too high for the firms most exposed.

If analysts under-react to geopolitical shocks, prices that rely on their forecasts may adjust slowly. Investors under-react to news about economically linked firms (Cohen and Frazzini, 2008), to oil price changes (Driesprong et al., 2008) and to news from the countries

where firms operate (Huang, 2015), and analysts' expectation errors line up with return predictability (Bouchaud et al., 2019; Engelberg et al., 2020). The results point to a specific channel through which geopolitical risk could reach prices with a delay, namely the forecasts of the analysts who cover exposed firms. They also show that forecasts are most likely to lag what is already public for firms whose 10-K shows sales or operations in the shocked country, and before those firms report.

More broadly, the same design applies wherever a firm's filings locate its exposure to a dated, public event whose consequences differ across firms in predictable ways. Disruptions at suppliers are a natural next application. Unlike a geopolitical shock to a country a firm buys from, a disruption at a firm's suppliers predictably lowers its output (Barrot and Sauvagnat, 2016; Boehm et al., 2019). Firms disclose supply-chain ties less fully than where they sell, and suppliers often withhold even the identities of their major customers (Ellis et al., 2012). Measuring how much of such a disruption analysts incorporate would show whether the under-reaction documented here extends from the markets where firms sell to the suppliers they depend on.

In the end, geopolitical shocks are public, dated and large, and a firm's 10-K tells analysts where it sells. That analysts still underestimate their predictable consequences for the better part of a year suggests that translating geopolitics into earnings is hard even for the professionals whose task it is.

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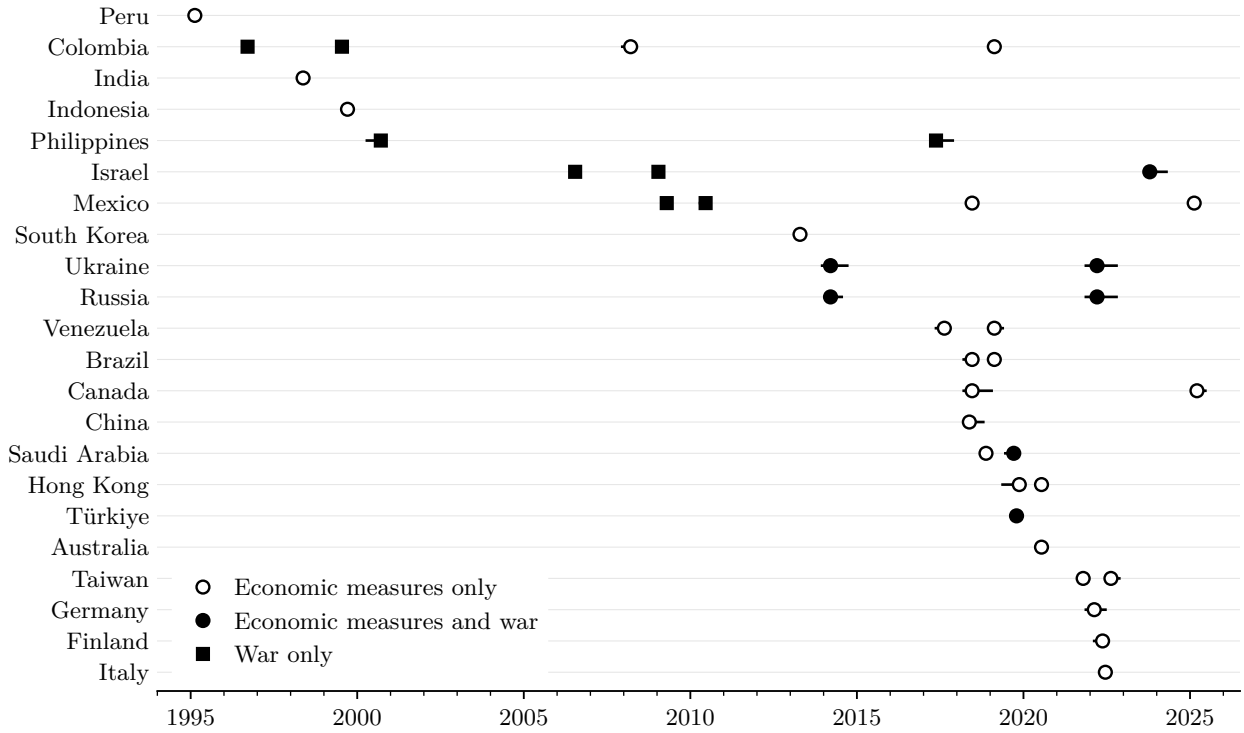
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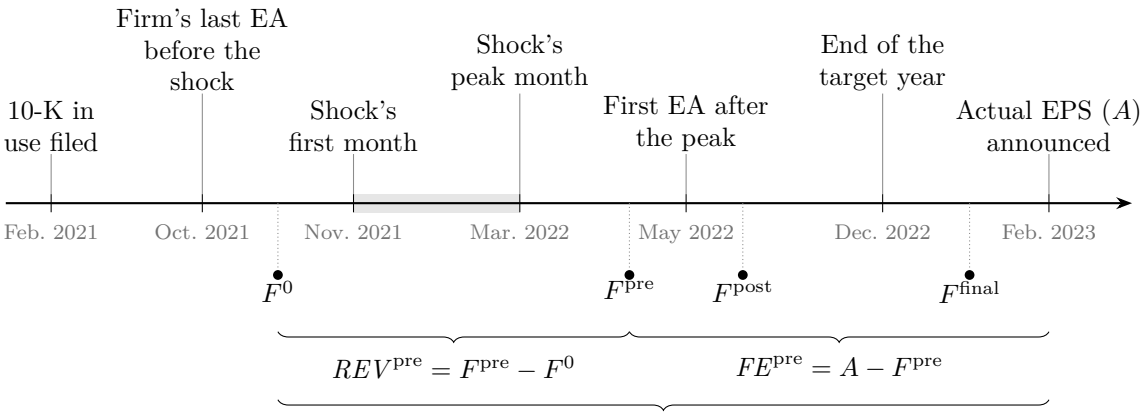
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Figure 1: The Geopolitical Shocks, 1995–2025



Each row is one of the 22 shocked countries, ordered by its first shock. For each of the 39 shocks a line runs from its first to its last month, and the marker sits at its peak month: open circles for economic measures only (24), filled circles for economic measures and war (7), and filled squares for war only (8). Markers aligned across rows are shocks of one event in several countries. Table 1 lists the shocks.

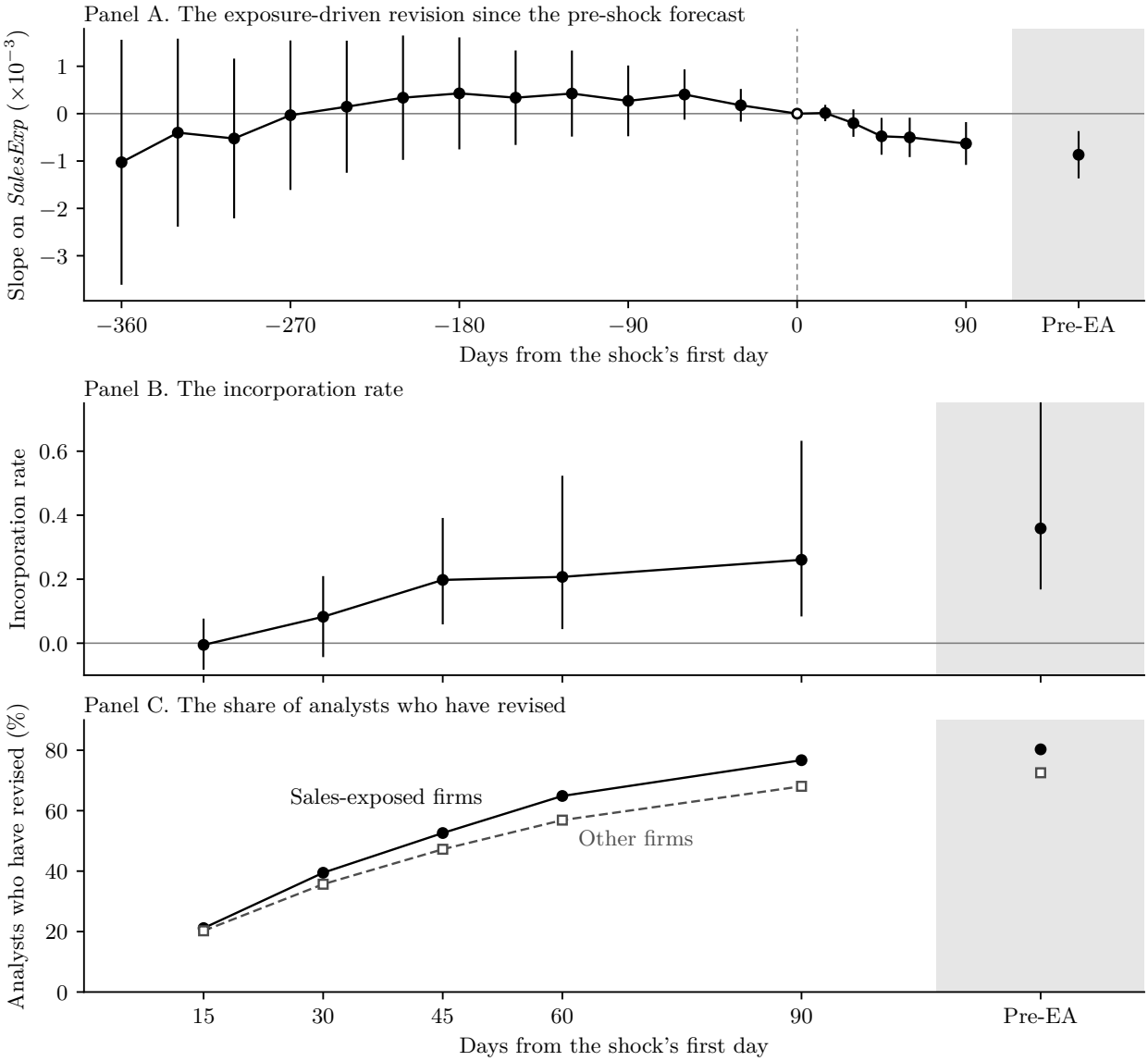
Figure 2: The Timing of Forecasts around a Shock



$FE^0 = A - F^0 = REV^{\text{pre}} + FE^{\text{pre}}$, each scaled by the share price before the shock

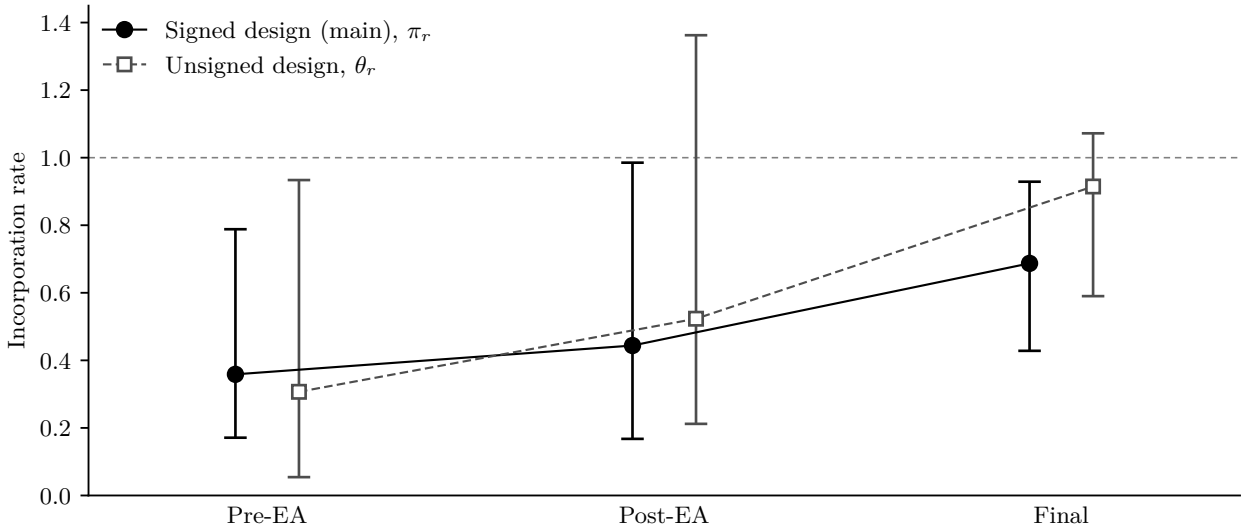
The events and forecasts that define each observation (not to scale), with the dates of one example: the 2022 shock to Russia, which began in November 2021 and peaked in March 2022 (Table 1), for a firm with a December fiscal year. The 10-K in use is the firm's latest 10-K filed before the shock's first month, at most 18 months old. The pre-shock forecast F^0 is the analyst's last forecast before the first month, issued or confirmed after the firm's last earnings announcement (EA) before the shock. The pre-EA forecast F^{pre} is in effect on the day before the firm's first EA after the peak month, a median of 46 days after the end of the peak month; the post-EA forecast F^{post} is the first forecast on or after that EA; and the final forecast F^{final} is the last forecast before the target year's annual earnings announcement. The target year is the first fiscal year ending at least six months after the peak month. The same split holds at each post-shock forecast r: $FE^0 = REV^r + FE^r$. The shaded band runs from the shock's first month to its peak month.

Figure 3: When Analysts Revise



Days are counted from the first day of the shock's first month (dashed line in Panel A); each analyst's forecast on a day is the one in effect then, the latest dated before it, or the pre-EA forecast if the firm announces earlier, and the pre-EA forecast is shown apart (right, shaded). Panel A: the slope on sales exposure of each analyst's forecast in effect on each day minus the pre-shock forecast F^0, in thousandths of the share price, with its 95% interval, from day −360 to the pre-EA forecast, where it is Table 4's reduced form; it is zero on the shock's first day (hollow circle), and each day uses the analyst-firm-shocks with a forecast that day (49% at day −360, all from day 0). Before the shock's first day, a positive value means that forecasts were cut between that day and F^0. Panels B and C, on days 15, 30, 45, 60 and 90: Panel B gives the IV estimate of the incorporation rate on each day, with its 95% Anderson–Rubin set, from the specification of Table 4, column (3), whose first stage it uses on every day, so only the revision changes. At the pre-EA forecast it is that table's π_{pre}. Panel C gives the share of analysts whose forecast in effect is dated on or after the first day of the shock's first month, weighted as in the regressions, for analysts of firms whose 10-K mentions selling to the shocked country (filled circles) and of other firms (hollow squares). By the pre-EA forecast, 21% of analyst-firm-shocks (26% weighted) have not revised; the analysts who have revised incorporate 0.48 $[0.20, 2.22]$ (effective first-stage $F = 6.32$). Standard errors are clustered by shock (39 clusters).

Figure 4: The Incorporation Rate after the Shock



The IV estimates of the incorporation rate at the pre-EA, post-EA and final forecasts, with their 95% Anderson–Rubin sets. Filled circles: π_r, the signed design (Table 4, Panel B). Hollow squares: θ_r, the unsigned design (Table 7, Panel B). Because it is measured in absolute errors, θ_r is not the same quantity as the signed rate. Before the firm reports, it is at least as large as the signed rate of the same firms, so there it can only overstate incorporation (Appendix A.4). The two series are offset to either side of each forecast for legibility. The dashed line marks full incorporation, a rate of one. Street EPS, 39 shocks.

Table 1: The Geopolitical Shocks

Country	Event	Type	Start	Peak	Peak z
Peru	Cenepa War between Peru and Ecuador, 1995	Measures	Feb 1995	Feb 1995	3.9
Colombia	Colombian armed conflict, 1964–	War	Sep 1996	Sep 1996	5.0
India	Indian and Pakistani nuclear tests, 1998	Measures	May 1998	May 1998	18.4
Colombia	Colombian armed conflict, 1964–	War	Jul 1999	Jul 1999	4.4
Indonesia	East Timorese crisis, 1999	Measures	Sep 1999	Sep 1999	14.1
Philippines	Moro conflict, 1968–2019	War	Apr 2000	Sep 2000	5.1
Israel	Lebanon War, 2006	War	Jul 2006	Jul 2006	5.6
Colombia	Andean diplomatic crisis, 2008	Measures	Dec 2007	Mar 2008	16.6
Israel	Gaza War, 2008–2009	War	Jan 2009	Jan 2009	5.5
Mexico	Mexican drug war, 2006–	War	Mar 2009	Apr 2009	4.1
Mexico	Mexican drug war, 2006–	War	Apr 2010	Jun 2010	4.7
South Korea	Korean crisis, 2013	Measures	Mar 2013	Apr 2013	4.1
Ukraine	Russo-Ukrainian War, 2014–	Both	Dec 2013	Mar 2014	27.2
Russia	Russo-Ukrainian War, 2014–	Both	Mar 2014	Mar 2014	7.3
Philippines	Moro conflict, 1968–2019	War	May 2017	May 2017	6.0
Venezuela	Venezuelan constitutional crisis, 2017	Measures	May 2017	Aug 2017	8.1
Brazil	U.S. steel and aluminum tariffs on allies, 2018	Measures	Mar 2018	Jun 2018	3.2
Canada	U.S. steel and aluminum tariffs on allies, 2018	Measures	Mar 2018	Jun 2018	7.8
China	U.S.–China trade war, 2018–2020	Measures	Apr 2018	May 2018	3.5
Mexico	U.S. steel and aluminum tariffs on allies, 2018	Measures	Jun 2018	Jun 2018	4.1
Saudi Arabia	Killing of Jamal Khashoggi, 2018	Measures	Oct 2018	Nov 2018	3.4
Brazil	Venezuelan presidential crisis, 2019–2023	Measures	Jan 2019	Feb 2019	5.6
Colombia	Venezuelan presidential crisis, 2019–2023	Measures	Jan 2019	Feb 2019	9.2
Venezuela	Venezuelan presidential crisis, 2019–2023	Measures	Jan 2019	Feb 2019	8.1
Hong Kong	Hong Kong protests, 2019–2020	Measures	May 2019	Nov 2019	5.8
Saudi Arabia	Yemeni civil war, 2014–	Both	Jun 2019	Sep 2019	4.2
Türkiye	Syrian civil war, 2011–2024	Both	Oct 2019	Oct 2019	5.4
Australia	Australia–China tensions, 2020–2021	Measures	May 2020	Jul 2020	6.5
Hong Kong	Hong Kong national security law, 2020	Measures	May 2020	Jul 2020	5.9
Taiwan	Chinese military pressure on Taiwan, 2021	Measures	Sep 2021	Oct 2021	4.4
Germany	Russo-Ukrainian War, 2014–	Measures	Nov 2021	Feb 2022	4.8
Russia	Russo-Ukrainian War, 2014–	Both	Nov 2021	Mar 2022	5.2
Ukraine	Russo-Ukrainian War, 2014–	Both	Nov 2021	Mar 2022	6.1
Finland	Finland's accession to NATO, 2022–2023	Measures	Feb 2022	May 2022	5.2
Italy	Russo-Ukrainian War, 2014–	Measures	Jun 2022	Jun 2022	3.3
Taiwan	Chinese military pressure on Taiwan, 2022	Measures	Aug 2022	Aug 2022	5.2
Israel	Gaza War, 2023–2025	Both	Oct 2023	Oct 2023	15.6
Mexico	U.S. tariffs on Canada and Mexico, 2025–2026	Measures	Jan 2025	Feb 2025	5.7
Canada	U.S. tariffs on Canada and Mexico, 2025–2026	Measures	Feb 2025	Mar 2025	5.3

The 39 shocks, from 26 events, in order of their first month. A shock is an episode of elevated geopolitical risk in one country. A month is elevated when the country's share of the Caldara–Iacoviello geopolitical-risk articles exceeds its mean over the previous 60 months by more than two standard deviations; elevated months separated by at most two other months form one episode. Each shock peaks at a z-score of at least three, involves the country directly, and brought economic measures against the country (Measures), war on or against its territory, with at least 1,000 battle-related deaths in a calendar year or a large-scale invasion (War), or both, known by the end of the peak month. Start is the first month, Peak the month with the highest share, and Peak z its z-score, (share − mean over the previous 60 months) / standard deviation.

Table 2: What Firms Say: Examples of Country Mentions in 10-Ks

Group	Firm, shock and source	Passage
Sales	Apple (China, 2018) 10-K of Nov. 3, 2017 Item 7 (MD&A)	Greater **China** net *sales* decreased during 2017 compared to 2016 due primarily to lower net sales of iPhone, partially offset by higher net sales of Services.
Sourcing	Lowe's (China, 2018) 10-K of Apr. 4, 2017 Item 1A (Risk Factors)	We source a large number of our products from foreign manufacturers with **China** continuing to be the dominant *import* source.
Operations	Intuitive Surgical (Mexico, 2025) 10-K of Jan. 31, 2024 Item 1 (Business)	We *manufacture* our instruments at our facilities in Sunnyvale, California and Mexicali, **Mexico**.
Not counted	Salesforce (Russia, 2022) 10-K of Mar. 17, 2021 Item 1A (Risk Factors)	Certain countries outside of the EEA (e.g., **Russia**, China and India) have also passed or are considering passing laws requiring varying degrees of local data residency.

One mention of the shocked country per row, from the 10-K in use of a firm in the signed design's main sample for that shock. A mention belongs to a group when one of the group's words appears within 25 words of it in the same paragraph (Section 3.1); each mention here belongs to its row's group only. Not counted: no business word in the window, so the mention counts only in total exposure. Bold: the country term; italics: the business word. Quotes are verbatim from the 10-K. Appendix Table A6 gives further examples and the selection rules.

Table 3: Descriptive Statistics

Panel A. Sample

Shocks	Firm-shocks	Firms	Analysts	Analyst-firm-shocks	Analysts per firm-shock	Sales-exposed (%)
39	29,177	2,824	7,761	268,865	9.2	16.5

Panel B. Exposure

	N	Mean	SD	P25	Median	P75	P90
Mentions per 10,000 words where positive: all, and near words of each group							
All	7,554	1.87	4.11	0.28	0.71	1.84	4.19
Sales words	4,811	1.02	1.62	0.23	0.48	1.08	2.42
Sourcing words	928	0.44	0.48	0.18	0.26	0.48	0.87
Operations words	3,996	0.98	1.56	0.23	0.46	1.06	2.19
Exposure, $\ln(1+x)$*, all firm-shocks*							
TotalExp	29,177	0.19	0.47	0.00	0.00	0.12	0.73
SalesExp	29,177	0.09	0.28	0.00	0.00	0.00	0.30
SourceExp	29,177	0.01	0.07	0.00	0.00	0.00	0.00
OpsExp	29,177	0.07	0.25	0.00	0.00	0.00	0.22

Panel C. Forecast errors and revisions, thousandths of the share price

	N	Mean	SD	P25	Median	P75	P90
FE^0	268,865	−2.7	40.1	−9.3	0.0	5.8	19.6
REV^{pre}	268,865	−1.6	15.6	−2.7	0.0	1.1	5.9
REV^{post}	251,124	−2.1	20.9	−5.4	−0.3	2.7	9.9
REV^{final}	268,864	−3.1	35.2	−8.5	−0.2	4.2	16.0
FE^{pre}	268,865	−1.1	34.6	−6.4	0.5	5.4	16.9
FE^{post}	251,124	−0.5	30.9	−4.4	0.6	4.5	14.3
FE^{final}	268,864	0.4	16.4	−0.5	0.5	2.2	6.3

Panel D. Controls (firm-shocks)

	N	Mean	SD	P25	Median	P75	P90
Log market value	29,177	8.86	1.35	7.89	8.63	9.68	10.71
Countries mentioned	29,177	12.6	10.6	4.0	11.0	18.0	26.0
ln(1 + foreign mentions)	29,177	2.44	1.18	1.71	2.70	3.29	3.76
Log number of analysts	29,177	1.99	0.74	1.61	2.08	2.48	2.83
Months to the end of the target year	29,177	13.5	3.9	10.0	13.0	16.0	19.9
Days from peak month to pre-EA forecast	29,177	46.1	27.1	25.0	46.0	65.0	85.0
ROA	29,177	0.040	0.118	0.017	0.051	0.092	0.141
Leverage	29,177	0.30	0.21	0.16	0.30	0.42	0.56
Book-to-market	29,177	0.34	0.34	0.13	0.27	0.47	0.72
Loss	29,177	0.18	0.38	0.00	0.00	0.00	1.00
Earnings volatility	29,177	0.055	0.114	0.014	0.026	0.053	0.109
Past 12-month return	29,177	0.21	0.52	−0.09	0.13	0.39	0.75

The signed design's sample: analyst-firm-shocks for street EPS in large, clean, non-financial firms (Appendix Table A1 defines them and every variable). Sales-exposed: the 10-K mentions the shocked country near a sales word. Panel B: mentions per 10,000 words of the 10-K in use where positive, and the exposure variables over all firm-shocks. Panel C: FE^0, and REV^r and FE^r at the three post-shock forecasts, each with its own analysts; FE^0 and the revisions are winsorized at the 1st and 99th percentiles within each shock, and $FE^r = FE^0 - REV^r$. Panel D: the controls, once per firm-shock. Statistics are unweighted.

Table 4: The Signed Design: Forecast Errors, Revisions and the Incorporation Rate

Panel A. Exposure columns, pre-EA forecast

	TotalExp (1)	*SalesExp* (2)	*SalesExp* with *SourceExp*, *OpsExp* (3)
First stage: FE^0	0.18	−0.46	−2.42***
	(0.38)	(−0.86)	(−3.49)
Reduced form: REV^{pre}	−0.08	−0.20	−0.87***
	(−0.42)	(−1.29)	(−3.67)
FE^{pre} (first stage − reduced form)	0.25	−0.26	−1.55**
	(0.67)	(−0.52)	(−2.39)
IV estimate: π_{pre}	−0.42	0.44	0.36
	$(-\infty, \infty)$	$(-\infty, \infty)$	$[0.17, 0.79]$
Effective first-stage F	0.15	0.73	12.21
SourceExp: first stage			−1.64 (−0.56)
SourceExp: reduced form			0.40 (0.28)
OpsExp: first stage			3.39*** (3.64)
OpsExp: reduced form			0.99** (2.26)
Firm-shocks	29,177	29,177	29,177

Panel B. Column (3) at the three post-shock forecasts

	Pre-EA forecast	Post-EA forecast	Final forecast
First stage: FE^0	−2.42***	−2.34***	−2.42***
	(−3.49)	(−3.39)	(−3.49)
Reduced form: REV^r	−0.87***	−1.04***	−1.66***
	(−3.67)	(−2.98)	(−3.05)
FE^r (first stage − reduced form)	−1.55**	−1.30**	−0.76**
	(−2.39)	(−2.05)	(−2.38)
IV estimate: π_r	0.36	0.44	0.69
	$[0.17, 0.79]$	$[0.17, 0.99]$	$[0.43, 0.93]$
OLS: REV^r on FE^0	0.18***	0.32***	0.77***
	(9.72)	(10.25)	(60.72)
Effective first-stage F	12.21	11.49	12.21
Firm-shocks	29,177	28,985	29,177
Analyst-firm-shocks	268,865	251,124	268,864

Signed design (main). Each observation is an analyst-firm-shock (street EPS). FE^0 is actual EPS minus the analyst's pre-shock forecast, REV^r the post-shock forecast r minus the pre-shock forecast, and $FE^r = FE^0 - REV^r$, each over the share price before the shock. The first stage regresses FE^0, and the reduced form REV^r, on the column's instrument (*TotalExp* in column (1), *SalesExp* in (2) and (3)), with *SourceExp* and *OpsExp* in column (3) (slopes shown, t beside), the six standard and six firm controls, and industry (Fama–French 48) × shock fixed effects; each shock counts equally, and each firm equally within it. Slopes are in thousandths of the share price per unit of exposure; t-statistics in parentheses use standard errors clustered by shock. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G-1$ degrees of freedom ($G = 39$ shocks). The IV estimate π_r is the 2SLS coefficient of REV^r on FE^0, with the instrument excluded and the other regressors included: the reduced-form slope over the first-stage slope, with its 95% Anderson–Rubin set in brackets. With one instrument, the effective first-stage F (Montiel Olea and Pflueger, 2013) equals the clustered first-stage F. The OLS row is the same regression without the instrument. Panel B repeats column (3) at the three post-shock forecasts. Sample: large, clean, non-financial firms (Table 3). An Anderson–Rubin set with an infinite end is printed with a parenthesis at that end.

Table 5: The Same Firms in the Years before the Shock

	Shock year	Year before	Two years before	Shock year minus the mean of the two years before
Panel A. Pre-EA forecast				
First stage: FE^0	-3.35^{***}	-0.05	-1.51	-2.57^{**}
	(-4.18)	(-0.04)	(-1.39)	(-2.45)
Reduced form: REV^{pre}	-1.26^{***}	0.29	-0.85	-0.98^{**}
	(-4.67)	(0.67)	(-1.54)	(-2.53)
FE^{pre} (first stage − reduced form)	-2.09^{**}			-1.59
	(-2.57)			(-1.48)
IV estimate: π_{pre}	0.38			0.38
	$[0.20, 0.76]$			$[0.08, 2.12]$
Effective first-stage F	17.49			6.03
Firm-shocks	25,994			
Panel B. Final forecast				
First stage: FE^0	-3.32^{***}	0.04	-1.61	-2.53^{**}
	(-4.26)	(0.03)	(-1.52)	(-2.47)
Reduced form: REV^{final}	-3.07^{***}	0.21	-1.64	-2.35^{**}
	(-4.73)	(0.19)	(-1.62)	(-2.42)
FE^{final} (first stage − reduced form)	-0.25			-0.18
	(-1.02)			(-0.68)
IV estimate: π_{final}	0.92			0.93
	$[0.81, 1.11]$			$[0.60, 1.30]$
Effective first-stage F	18.14			6.11
Firm-shocks	25,978			

Table 4, column (3), at the firm level from the I/B/E/S consensus $\bar{F}$, for the same firms in the shock year and in each of the two fiscal years before it: the large, clean, non-financial firm-shocks with consensus forecasts and actuals in all three years; each panel's count applies to every column. Each year's consensus is taken the same number of years before the shock's dates and uses its own actual and share price: FE^0 is the actual minus the pre-shock consensus, REV^r the consensus at forecast r minus the pre-shock consensus, and $FE^r = FE^0 - REV^r$. The last column's variables are the shock year's minus the mean of the two earlier years'. As in Table 4, *SalesExp* is the instrument, and *SourceExp*, *OpsExp*, the controls and industry × shock fixed effects are included regressors. Slopes are in thousandths of the share price per unit of exposure; t-statistics in parentheses use standard errors clustered by shock. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G-1$ degrees of freedom ($G = 39$ shocks). The IV estimate π_r, the reduced-form slope over the first-stage slope, has its 95% Anderson–Rubin set in brackets.

Table 6: Alternative Inference and Placebo Countries

	Signed design (main)			Unsigned design	
	FE^0	REV^{pre}	FE^{pre}	ΔAFE	ΔIMP^{pre}
Slope on *SalesExp*	-2.42^{***}	-0.87^{***}	-1.55^{**}	1.93^{***}	0.59^{**}
	(-3.49)	(-3.67)	(-2.39)	(3.10)	(2.43)
Panel A. p-values					
Clustered t	0.001	0.001	0.022	0.004	0.020
Wild cluster bootstrap	0.001	0.003	0.020	0.002	0.024
Randomization, on t	0.002	0.002	0.016	< 0.001	0.022
Randomization, on the slope	0.025	0.022	0.097	0.008	0.116
Panel B. 20 placebo countries					
As strong, same direction	0	0	0	0	1
As strong, either direction	1	0	3	0	2
Strongest in the same direction: t	-2.37	-1.78	-2.10	2.41	2.53
	China	Israel	China	France	France
As strong on every variable, same direction		0		0	
As strong on every variable, either direction		0		0	

Signed design (main): Table 4, column (3); Unsigned design: Table 7, column (2); both at the pre-EA forecast. The slopes are in thousandths of the share price, with their t-statistics in parentheses. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G - 1$ degrees of freedom ($G = 39$ shocks). Panel A gives the p-values of the clustered t-statistic; of the restricted wild cluster bootstrap (t-statistic, Webb weights, 9,999 draws); and of randomization inference, with the firm-shocks' exposure variables shuffled together within industry $\times$ shock cells (1,000 draws), comparing the absolute t-statistic or the absolute slope with the actual one. Panel B replaces the shocked country's mentions with those of each of 20 other countries, dropping that country's own shocks. A placebo country is as strong when the absolute value of its t-statistic is at least the shocked country's, in the same direction if it also has the same sign. Appendix Table A7 lists every country.

Table 7: The Unsigned Design

Panel A. Exposure columns, pre-EA forecast

	TotalExp (1)	*SalesExp* (2)	*SalesExp* with *SourceExp*, *OpsExp* (3)
First stage: ΔAFE	1.38^{***}	1.93^{***}	2.24^{*}
	(3.36)	(3.10)	(1.69)
Reduced form: ΔIMP^{pre}	0.39^{*}	0.59^{**}	0.92^{**}
	(1.89)	(2.43)	(2.39)
IV estimate: θ_{pre}	0.28	0.31	0.41
	$[-0.03, 0.69]$	$[0.05, 0.93]$	$(-\infty, -1.21] \cup [0.09, \infty)$
Effective first-stage F	11.28	9.61	2.85
Firm-shocks	36,977	36,977	36,977

Panel B. Column (2) at the three post-shock forecasts

	Pre-EA forecast	Post-EA forecast	Final forecast
First stage: ΔAFE	1.93^{***}	1.92^{***}	1.92^{***}
	(3.10)	(3.07)	(3.00)
Reduced form: ΔIMP^{r}	0.59^{**}	1.00^{***}	1.75^{**}
	(2.43)	(3.19)	(2.64)
IV estimate: θ_r	0.31	0.52	0.91
	$[0.05, 0.93]$	$[0.21, 1.36]$	$[0.59, 1.07]$
OLS: ΔIMP^{r} on ΔAFE	0.20^{***}	0.33^{***}	0.90^{***}
	(7.86)	(9.06)	(43.14)
Effective first-stage F	9.61	9.45	8.99
Firm-shocks	36,977	37,020	36,885

Unsigned design. Each observation is a firm-shock, from the I/B/E/S consensus $\bar{F}$ (street EPS). $AFE^{0} = |A - \bar{F}^{0}|/P$ is the absolute pre-shock forecast error and $IMP^{r} = AFE^{0} - AFE^{r}$ the accuracy improvement by forecast r, for fiscal year t taken $|t|$ years before the shock's dates. ΔAFE and ΔIMP^{r} are the shock year minus the mean of the two earlier years (Appendix Table A10: other baselines). The first stage regresses ΔAFE, and the reduced form ΔIMP^{r}, on the column's instrument (as in Table 4), the six standard controls and industry × shock fixed effects; each shock counts equally, and each firm equally within it. Slopes are in thousandths of the share price per unit of exposure; t-statistics in parentheses use standard errors clustered by shock. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G - 1$ degrees of freedom ($G = 39$ shocks). The IV estimate θ_r, the 2SLS coefficient of ΔIMP^{r} on ΔAFE, has its 95% Anderson–Rubin set in brackets. OLS omits the instrument. Sample: large firms with no restatement or non-reliance filing, all industries (Appendix Table A1). Each forecast uses the firm-shocks with consensus forecasts at that forecast in all three years. An Anderson–Rubin set with an infinite end is printed with a parenthesis at that end.

Table 8: Exposure Counted in One Section of the 10-K, and Other Mentions

	Whole 10-K (1)	Item 7 (MD&A) (2)	Item 1 (Business) (3)	Item 1A (Risk Factors) (4)	Other mentions (5)
Panel A. Signed design (main), pre-EA forecast					
FE^0	-2.42^{***}	-1.60^{***}	-0.84	1.21	-1.24
	(-3.49)	(-2.99)	(-1.65)	(1.53)	(-1.09)
REV^{pre}	-0.87^{***}	-0.48^{***}	-0.37^{**}	0.35	-0.51
	(-3.67)	(-3.27)	(-2.28)	(1.03)	(-1.06)
FE^{pre}	-1.55^{**}	-1.12^{**}	-0.47	0.85	-0.74
	(-2.39)	(-2.20)	(-1.07)	(1.66)	(-0.95)
π_{pre}	0.36	0.30	0.44	0.29	0.41
	$[0.17, 0.79]$	$[0.12, 0.84]$	$(-\infty, -0.75] \cup [0.11, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$
Effective first-stage F	12.21	8.96	2.71	2.35	1.19
Placebos as strong, FE^0 / REV^{pre}:					
same direction	0 / 0	0 / 1	2 / 4	6 / 6	10 / 6
Firm-shocks	29,177	28,629	28,165	25,757	29,177
Panel B. Unsigned design, pre-EA forecast					
ΔAFE	1.93^{***}	1.12^{**}	1.17^{**}	0.35	1.88^{***}
	(3.10)	(2.16)	(2.19)	(0.41)	(3.39)
ΔIMP^{pre}	0.59^{**}	0.49^{**}	0.24	0.24	0.52
	(2.43)	(2.66)	(0.94)	(0.84)	(1.51)
θ_{pre}	0.31	0.44	0.20	0.70	0.28
	$[0.05, 0.93]$	$[0.13, 4.91]$	$[-1.09, 0.93]$	$(-\infty, \infty)$	$[-0.14, 0.68]$
Effective first-stage F	9.61	4.68	4.80	0.17	11.48
Placebos as strong, ΔAFE / ΔIMP^{pre}:					
same direction	0 / 1	3 / 0	1 / 3	6 / 3	0 / 1
Firm-shocks	36,977	36,241	35,718	33,717	36,977

Both designs with the shocked country's mentions counted in one part of the 10-K and scaled by that part's words: the whole 10-K (the main measure), Item 7 (MD&A), Item 1 (Business) or Item 1A (Risk Factors). Panel A is Table 4, column (3), and Panel B Table 7, column (2), with the section's exposure variables in place of the whole 10-K's. Each column covers the firm-shocks whose 10-K has that section (the firm-shock rows); Item 1A counts only in 10-Ks filed from December 1, 2005. Column (5) replaces *SalesExp* with the shocked country's other mentions in the whole 10-K, those with no sales, sourcing or operations word nearby, in logs per 10,000 words; Panel A keeps *SourceExp* and *OpsExp*. Slopes are in thousandths of the share price per unit of exposure; t-statistics in parentheses use standard errors clustered by shock. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G - 1$ degrees of freedom; G is 39 for the other columns and 33 for Item 1A (Risk Factors). Incorporation rates have 95% Anderson–Rubin sets in brackets. Placebos as strong: of 20 placebo countries, those whose absolute t-statistic is at least the shocked country's, with the same sign, with the same section's mentions (Table 6), or, in column (5), with each placebo country's mentions of every kind in place of the shocked country's. An Anderson–Rubin set with an infinite end is printed with a parenthesis at that end.

Table 9: Good News and Bad News

Panel A. Slopes on OpsExp (good news)	Pre-EA forecast	Post-EA forecast	Final forecast
First stage: FE^0	3.39***	3.27***	3.39***
	(3.64)	(3.48)	(3.64)
Reduced form: REV^r	0.99**	1.20*	2.66***
	(2.26)	(1.85)	(3.29)
FE^r (first stage − reduced form)	2.40***	2.08***	0.73*
	(3.44)	(2.87)	(1.96)
Panel B. Incorporation rates			
IV estimate: π_r, bad news (*SalesExp*)	0.36	0.44	0.69
	[0.17, 0.79]	[0.17, 0.99]	[0.43, 0.93]
IV estimate: π_r, good news (*OpsExp*)	0.29	0.37	0.78
	[0.05, 0.50]	[−0.06, 0.71]	[0.55, 1.01]
Effective first-stage F	13.24	12.11	13.24
Difference, good minus bad news	−0.07	−0.08	0.10
p-value	0.556	0.639	0.236
Common rate, both instruments	0.30	0.38	0.77
	[0.13, 0.59]	[0.09, 0.85]	[0.44, 1.03]
Effective first-stage F	8.79	7.99	8.79
Firm-shocks	29,177	28,985	29,177
Analyst-firm-shocks	268,865	251,124	268,864

Signed design (main), Table 4, column (3), with each exposure in turn as the instrument; FE^0, REV^r and FE^r are as there. Panel A gives the slopes on operations exposure in the regressions of Table 4, column (3), which also include *SourceExp*, the six standard and six firm controls, and industry (Fama–French 48) × shock fixed effects; each shock counts equally, and each firm equally within it. Slopes are in thousandths of the share price per unit of exposure; t-statistics in parentheses use standard errors clustered by shock. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G - 1$ degrees of freedom ($G = 39$ shocks). Panel B gives the IV estimate of REV^r on FE^0 with *SalesExp* (bad news) or *OpsExp* (good news) as the excluded instrument and the other exposures included, with its 95% Anderson–Rubin set and, for good news, the effective first-stage F (Table 4 gives that of bad news); the difference between the two, with its p-value from the slopes' joint covariance (influence functions summed by shock); and the 2SLS rate with both exposures as instruments, with its two-instrument Anderson–Rubin set and the effective first-stage F (Montiel Olea and Pflueger, 2013). Before the firm reports, of 20 placebo countries none is as strong in the same direction on the operations slope of FE^0, none on that of FE^{pre} and three on that of REV^{pre}; their wild cluster bootstrap p-values are 0.001, 0.002 and 0.043. Against the mean of the two earlier years (Table 5), the operations slope of FE^0 has $t = 1.83$.

Table 10: Same-Country Experience

	Pre-EA forecast	Post-EA forecast	Final forecast
Panel A. Colleagues on the same firm in the same shock			
Additional incorporation rate	−0.01	−0.02	0.03
	[−0.21, 0.19]	[−0.20, 0.16]	[−0.25, 0.31]
t, FE^r	0.97	0.51	1.25
t, FE^0	1.02	0.57	1.02
Amount, any country (per SD)	0.00	−0.01	−0.04
	[−0.12, 0.11]	[−0.14, 0.13]	[−0.22, 0.15]
Large broker (top decile)	−0.03	−0.08	−0.02
	[−0.17, 0.12]	[−0.28, 0.13]	[−0.24, 0.21]
Analyst-firm-shocks	268,865	251,124	268,864
Panel B. The same analyst across shocks			
Additional incorporation rate	0.16	0.17	0.36
	[−0.09, 0.41]	[−0.12, 0.46]	[0.01, 0.71]
t, FE^r	0.29	−0.20	2.05
t, FE^0	−0.73	−1.17	−0.73
Analysts seen with and without same-country experience	2,206	2,111	2,206
at sales-exposed firms	1,735	1,656	1,735
Analyst-firm-shocks	265,816	248,097	265,815

Same-country experience (Appendix Table A1): during an earlier, finished episode of economic measures or war that directly involved the same country (one of 102), the analyst forecast a firm whose 10-K mentioned selling there, with its annual earnings out before the new shock. Both panels add experience, its interactions with the three exposures and firm × shock fixed effects to Table 4, column (3); Panel B adds each analyst's intercept and exposure slopes, dropping rows these fit exactly. Panel A's amount row uses the standardized amount in any country (Appendix Table A1, "Amount"), and its large-broker row compares analysts at brokers in the top decile by number of analysts in the target fiscal year, across all U.S. I/B/E/S coverage, with their colleagues on the same firm. The additional incorporation rate is the slope of experience × *SalesExp* on REV^r over Table 4's first stage at that forecast, with its 95% confidence interval in brackets and the t-statistics of the same interaction's slopes on FE^r and FE^0. Standard errors are clustered by shock (39 clusters). Checks of Panel B's final rate: 2 of 16 placebo experiences and 0 of 20 placebo exposures are as strong in the same direction; leaving out each shock, 18 of 39 estimates are significant at 5 percent (0.26 to 0.44); shuffling experience within firm × shock (1,000 draws) gives $p = 0.072$.

A Online Appendix

This appendix has four parts. Appendix A.1 defines the variables and traces the sample. Appendix A.2 describes how the 39 shocks are built from the geopolitical risk index and episode labels. Appendix A.3 describes the text measure of sales exposure. Appendix A.4 reports the additional results to which the main text refers.

A.1 Variable definitions and sample selection

Table A1 defines every variable and gives its source. It follows the order of the paper: the forecast errors and revisions of the signed design, the absolute forecast errors and accuracy improvements of the unsigned design, the exposure measures, the controls, the sample filters and the experience measures of Section 7.

Per-share quantities are treated the same way throughout. Each is divided by the share price at the end of the month before the shock's first month, and the tables report slopes in thousandths of that price. The pre-shock forecast error and each forecast revision are winsorized at the 1st and 99th percentiles within each shock. Each post-shock forecast error is then their difference, so that $FE^0 = REV^r + FE^r$ holds exactly for every observation.

Table A2 traces both designs' samples step by step. Of the 248,504 firm-shocks with a 10-K in use in the 39 shocks, 152,999 are linked to I/B/E/S and CRSP, and 101,264 keep at least one analyst whose forecasts pass the rules of Section 3.2. Most of the analysts who leave do so because their last forecast before the shock was neither issued nor confirmed after the firm's last earnings announcement. In the signed design, the two largest cuts are the financial industries and the restriction to large firms, which by construction keeps about half of the firm-shocks left at that step; 29,177 firm-shocks and 268,865 analyst-firm-shocks remain. The unsigned design keeps financial firms and firms with large special items but needs the same firm's consensus forecasts in the two earlier years, and 36,977 firm-shocks remain.

A.2 Shock construction

The shocks are built in three steps: the country indexes date episodes of elevated geopolitical risk, each episode is coded, and four rules select the shocks. Table A3 summarizes the coding scheme and the selection, and Table A4 lists the candidates left out.

A.2.1 The index and the episodes

The country indexes count the articles in ten American, British and Canadian newspapers that discuss geopolitical risk and mention a given country or its major cities, as a percent of all articles, for 44 countries (Caldara et al., 2026). I use the download of September 4, 2026, which runs from January 1985 to August 2026 (Caldara and Iacoviello, 2026). Dividing each country's index by the percent of all articles that discuss geopolitical risk gives the country's share of geopolitical-risk news. The share rises only when a country becomes a larger part of that news. In September 2001, when geopolitical news surged everywhere, Sweden's index rose to 6 to 16 times its summer values while its share stayed near 1 percent; in March 2014, 59 percent of geopolitical-risk articles mentioned Ukraine, up from 5 percent in January. Figure A1 shows the Swedish case.

These shares date the episodes. A month is an elevated month for a country when its share exceeds the mean of the previous 60 months by more than two standard deviations of those months and at least ten geopolitical-risk articles mention the country. Elevated months separated by at most two other months form one episode, with a first, a peak and a last month. The peak month is the episode's elevated month with the highest share, and the episode's peak z-score is the z-score in that month. From January 1990, the rule finds 659 episodes, including 18 of 19 well-known events checked against it; episodes of the United States are not used. Dating episodes at two standard deviations places the pre-shock forecast before the build-up: the 2022 shock to Russia starts in November 2021 and peaks in March 2022. An episode is selected only if its peak reaches three standard deviations. Among the 39 shocks, the peak z-score ranges from 3.2 (Brazil, 2018) to 27.2 (Ukraine, 2014). Figure A2 shows the rule at work in each of the 22 shocked countries.

A.2.2 Coding and the two definitions

I hand-code each of the 651 episodes outside the United States, checking every label against its sources (Table A3, Panels A and B). Each label names the event most responsible for the country's coverage in the peak month, the types of act in which the country itself took part, whether it was directly involved, its role and its counterparts. The type definitions adapt those of established datasets, such as the UCDP/PRIO definition of armed conflict (Gleditsch et al., 2002) and the Global Sanctions Data Base (Felbermayr et al., 2020). Each label rests on at least three reputable sources that report the event within the episode's months; a later evidence check recorded 2,868 sources.

Two definitions decide which episodes qualify. Economic measures are coercive steps actually imposed by a state or a group of states that restrict what private parties may

trade or finance with another economy, or that cut state-controlled energy. They comprise embargoes, import and export bans, export controls, coercive tariffs, energy cuts, closures of borders, ports or airspace to the other state's trade, and financial sanctions on the state, its central bank, its banks or its major firms. Aid suspensions and sanctions on named individuals count as diplomatic measures instead, because they rarely restrict the trade or finance through which a shock reaches firms. War is fighting in a conflict with at least 1,000 battle-related deaths in a calendar year of the episode, by UCDP's figures (Davies et al., 2026), or a large-scale invasion of another state. This is UCDP's war threshold (Gleditsch et al., 2002); the Correlates of War project uses 1,000 combatant deaths within a twelve-month period (Sarkees and Wayman, 2010).

A.2.3 Selection and the independent check

Table A3, Panel C, shows how the episodes are narrowed to the shocks. Of the 377 episodes in which the country is directly involved, 102 involve economic measures or war, and 66 of these reach three standard deviations at the peak. Two judgments, made episode by episode with a one-line reason, then remove 25. The first is that the country must be on the receiving end, meaning that its economy must take the hit through those measures or that war. Countries at which measures were aimed qualify, including through retaliation, such as Canada and Mexico in 2018 and 2025, and so does Russia in 2014 and 2022, which fought abroad but faced sanctions; countries that only imposed measures, or whose forces acted only abroad, do not. The second is that the qualifying measure or war must be public by the end of the peak month, so that analysts could know it when they issued their pre-EA forecasts, with announced measures counting from their announcement. Saudi Arabia in 1990 is out under this rule, because fighting on its soil began only in January 1991, while Mexico in 2025 qualifies through the tariff order of February 1, which took effect on March 4.

Two more episodes, in Saudi Arabia in March and July 2026, are set aside because their target year has no actual EPS yet. Table A4 lists all 27 candidates left out, with their reasons. The remaining 39 shocks come from 26 events and qualify through export controls, financial sanctions, U.S. tariffs, other states' bans and border closures, Russian energy cuts and airspace closures, or war on the country's territory (Table 1).

A separate review checked all 66 candidates against the rules in chronological batches, naming each episode's qualifying measure or war, its date and direction, and a source. It led to two changes. Germany in 2021–2022 entered, because on February 28, 2022, the last day of its peak month, Russia closed its airspace to German airlines. Venezuela in 2025–2026 dropped out, because under the definitions neither the U.S. tanker blockade nor the raid on

Caracas was an economic measure or a war, and the economic measures in that episode were Venezuela's own. Each of the 39 shocks has a qualifying measure or war with a date and a source.

A.3 The text measure

Sales exposure is built in four steps: find each mention of a country in the 10-K, drop terms and phrases that do not refer to the country, sort the remaining mentions by the business words around them, and scale the count by the 10-K's length. This part describes each step, the choice of 10-K and section, the example passages and two comparisons with other data.

A.3.1 The 10-Ks and the dictionary

The text comes from a cleaned copy of 295,360 annual reports on Forms 10-K, 10-K405 and 10-KSB filed on EDGAR from July 1994 to July 2026. The copy holds the main 10-K document. Filers that incorporated their MD&A from an annual-report exhibit have no MD&A text in it: about 18 percent of filings in 1994–2000, 7 percent in 2001–2005 and 3 percent or fewer later.

A rule-based dictionary finds the country mentions. Its terms come from standard sources: ISO 3166 and UN country names, Unicode CLDR names, SEC EDGAR country names, the CIA World Factbook's names, nationalities and abbreviations, and Wikidata demonyms. General rules rather than term-by-term choices decide what counts. A term that the sources give to several countries counts for the only sovereign state among them: "British" counts for the United Kingdom and "Dutch" for the Netherlands. Constituent countries count for their parent, as England, Scotland and Wales count for the United Kingdom. A term identical to a U.S. state name does not count ("Georgia"), and U.S. state, census-region and foreign subdivision names that contain a country's term block it, as in "New Mexico", "New Jersey", "New England", "British Columbia" and "New South Wales". Names of seas, gulfs and straits likewise block the terms they contain, as in "Gulf of Mexico" and "South China Sea". Five stated exceptions assign "Korea" and "Korean(s)" to South Korea and "Republic of China" and "Chinese Taipei" to Taiwan. The dictionary has 1,162 country terms (545 names, 529 demonyms and 88 abbreviations) for 248 countries and territories, and 86 blocking phrases.

The dictionary counts a country's names, demonyms and standard abbreviations, but not its cities, subdivisions or regions. Firms name the countries they do business with when they discuss sales, segments and properties, so these terms capture most of the relevant disclosure. The dictionary finds 99 percent of the countries that firms report among their

Compustat geographic segments, and it finds 94.3 percent of the countries that Hoberg and Moon (2017) record as a firm's output countries. City names, by contrast, are often ambiguous in filings, because many are also U.S. towns, company names or ordinary words, and a region such as Europe does not identify the shocked country. Earnings-call measures such as Hassan et al. (2024) also draw on towns and subdivisions, because speakers often refer to a country through its places. Omitting cities and provinces leaves some exposure unmeasured: in hand-labeled 10-K passages, about 18 percent of real exposures name only a city or province. Such omissions are measurement error in exposure, which attenuates the first stage and the reduced form by the same factor and leaves the incorporation rate unchanged (Section 4.2).

Three conventions further limit what counts. Only capitalized names and demonyms count ("Turkey", not "turkey"), and abbreviations count only in capitals. A country name followed by a U.S. address does not count ("Lebanon, PA"). The filer's own name is masked, so that a company named after a country does not mention that country by naming itself.

A.3.2 The term audit

An audit checks the 200 dictionary terms that occur most often in 10-Ks. Its rule was fixed before any sampling. Each step draws its own random sample, one hit per 10-K, and classifies each hit as referring to the country or not. The first step screens each term on 30 hits and keeps the terms with at least 27 correct hits, 161 of the 200. For each of the other 39 terms, a second, discovery step draws 30 more hits and assigns each wrong hit the phrase that makes it wrong, and a phrase that matches at least three wrong hits and no correct one becomes a candidate blocking phrase. A third step validates each candidate on up to 30 new 10-Ks and adopts it only if every hit it matches is wrong. A final holdout step draws 60 new hits per remaining term, after the adopted blocking phrases, and excludes a term if fewer than half of these hits refer to the country.

The audit excludes 15 terms, among them "Jordan", "Jersey" and "Chad", which often refer to a person or a U.S. place, and abbreviations such as "CH", "NL" and "PNG". It adds 24 blocking phrases for 15 kept terms, such as "British thermal unit", "Dutch auction", "Indian tribe", "Inner Mongolia" and "Uruguay Round". Of 100 hits drawn at random from the audit's 10,494 classifications and classified again without the first result shown, all 100 agree. After the audit, 0.56 percent of the 10-K–country pairs with a mention of a country that has a risk index rest only on terms outside the audit. Table A5 shows what counts and what does not: every counted term of nine countries, eight of them shocked and the United Kingdom to show the constituent-country rule, and examples of each rule that keeps a mention from counting.

A.3.3 Business words and scaling

Following Hoberg and Moon (2017), a mention belongs to a group of business words when one of the group's words appears in the same paragraph, at most 25 words before or after it. Paragraphs end at blank lines, and a word is any token with at least one letter. The three groups take Hoberg and Moon's word lists unchanged. Sales words are their output words: sales, revenue(s), customer(s), consumer(s), market(s), marketing, distribution, distributor(s), dealer(s), client(s), export(s), shipments, demand(s), store(s), wholesale and receivable(s), with related forms. Sourcing words are their external-input words: supplier(s), import(s), subcontractor(s), vendor(s), and "purchase(s)" or "purchased" followed by "from" within three words. Operations words are their internal-input words (subsidiary, facility, plant, factory, venture, exploration, storage, warehouse) and their production words (manufacture, manufacturing, produce, production).

Across all 10-Ks, 28.6 percent of counted country mentions have a sales word nearby, 2.7 percent a sourcing word, 20.4 percent one of the operations group's internal-input words, such as facility or plant, and 8.9 percent one of its production words. In all, 47.4 percent have at least one business word. In a random sample of 30 mentions with a business word, 26 described the firm's business in the country.

A mention counts in every group whose words surround it, and in none if no business word is nearby. Mentions are recorded by their combination of groups, so that a mention near the words of two groups counts once in each and each group's count includes every mention with that group's words exactly once. Each count is divided by the words in the 10-K in use and multiplied by 10,000; numbers, punctuation and table padding do not count as words. A firm whose 10-K in use does not mention the shocked country next to a group's words has zero exposure for that group. Among sales-exposed firms in the main sample, the median is 0.48 sales mentions per 10,000 words, about one in a 20,000-word 10-K (Table 3).

A.3.4 The 10-K in use and its sections

Each month, a firm uses its latest 10-K filed before the month began, if that 10-K is at most 18 months old; 10-Ks filed the same day are ordered by their period of report. While an episode of elevated geopolitical risk (Appendix A.2), not only one of the 39 shocks, is under way in a country that the current or the new 10-K mentions, the firm keeps its current 10-K. It switches to its latest 10-K when those episodes end. A shock's exposure comes from the 10-K in use in the shock's first month.

A firm can keep its 10-K this way for any length of time, so the 10-K in use can be older than 18 months. Across all firm-months, 6.6 percent use such a kept 10-K. The 10-K in use

is a median of 7 months old and 13 months old at the 90th percentile, and it is older than 18 months in 2.0 percent of firm-months. Firms whose 10-K mentions many countries with a risk index keep their 10-K more often.

To measure exposure by section, a parser locates the 10-K's item headings and assigns each mention to the item in which it falls. Text before the first heading is front matter, and the items of Form 10-KSB are mapped to 10-K numbering. Where an independent parser, EdgarTools, also assigns an item, the two agree on 92.8 percent of mentions in a comparison of 857 filings. Across all 10-Ks, 30 percent of foreign-country mentions fall in Item 1, 15 percent in Item 7 and 13 percent in Item 1A. A section's exposure counts the mentions in that item per 10,000 words of the item.

Item 1A became the risk-factors item of Form 10-K with the Securities Offering Reform, which took effect on December 1, 2005, for fiscal years ending on or after that date (U.S. Securities and Exchange Commission, 2005). Earlier 10-Ks often used that number for other content, such as the list of executive officers. Item 1A therefore counts as risk factors only in 10-Ks filed on or after December 1, 2005. In the signed design's main sample of 29,177 firm-shocks, Item 1 is available for 28,165, Item 7 for 28,629 and Item 1A for 25,757 (Table 8).

A.3.5 Examples and comparisons with other data

Table A6 adds nine passages to the four of Table 2: two per group, one mention that counts in both the sales and the operations groups (Coca-Cola on Mexico), and two risk-factor mentions without a business word. Each comes from a well-known firm's 10-K in use for one of the shocks.

Two comparisons with other data support the measure. First, among firms that report Compustat segment sales in a country with a risk index, within each country and fiscal year with at least ten such firms (809 country-years in 35 countries), the Spearman correlation between the country's mentions per 10,000 words and its share of the firm's sales has a median of 0.45 and is positive in 98 percent of country-years. Second, for the latest 10-K filed in each year from 1997 to 2017, the business-word groups find 94.3 percent of the firm-year countries that Hoberg and Moon (2017) record as output countries and 94.2 percent of those they record as input countries.

A.4 Additional results

A.4.1 Placebo countries

Table A7 reports the t-statistics on sales exposure for each of the 20 placebo countries behind Table 6. Each row replaces the firm's exposure to the shocked country with its exposure to

the placebo country, counted the same way, and drops the placebo country's own shocks. In the signed design, the shocked country has the largest t-statistic in absolute value on the pre-EA revision (-3.67, against 2.65 for Switzerland, the next largest). On the pre-shock forecast error it has the second largest (-3.49), after the Netherlands, whose t-statistic has the opposite sign (4.60). On the pre-EA forecast error (-2.39), the Netherlands, Spain and Singapore have larger t-statistics, all with the opposite sign (5.17, 2.51 and 2.46).

In the unsigned design, France is the strongest placebo country on both outcomes ($t = 2.41$ on ΔAFE and 2.53 on ΔIMP^{pre}), and South Korea predicts a smaller accuracy improvement ($t = -2.86$). The shocked country has the largest t-statistic on ΔAFE (3.10), and no placebo country is as strong on both outcomes.

A.4.2 Leaving out shocks and overlap groups

Table A8, Panel A, drops each of the 39 shocks in turn. The rest of the sample is unchanged, because the market-value median, the winsorizing and the weights are all taken within each shock. In the signed design, π_{pre} ranges from 0.30 to 0.44 and π_{final} from 0.64 to 0.73, and every pre-EA set excludes zero. The rate is lowest without Canada in 2018, the steel and aluminum tariffs, and highest without Mexico in 2009, the drug war, whose own rate is close to zero. The effective first-stage F-statistic stays between 9.3 and 13.6. In the unsigned design, θ_{pre} ranges from 0.25 to 0.41, and every set excludes zero; the lowest, without Canada in 2025, is $\theta_{\text{pre}} = 0.25$ $[0.001, 0.84]$.

Some shocks run at the same time, such as those to Russia, Ukraine, Germany, Finland and Italy in 2021–2022. Overlap groups link shocks whose dates overlap, and the 39 shocks form 19 groups. Leaving out each group (Panel B) gives π_{pre} between 0.25 and 0.44, with the lowest value, 0.25 $[0.05, 0.67]$, without the eleven shocks of 2018–2019, which include the U.S.–China trade war. Clustering by the 19 groups (Panel C) leaves the estimates unchanged and widens the sets. In the signed design, the t-statistics are -3.22 for the pre-shock forecast error and -3.64 for the revision, and the set for π_{pre} is $[0.16, 0.93]$ (effective first-stage $F = 10.4$). In the unsigned design, the t-statistics are 2.35 and 2.31, and the set for θ_{pre} is $[0.04, 2.11]$ (effective first-stage $F = 5.5$).

A.4.3 Shock-level weights and shock types

With one instrument, the IV estimate is a weighted average of shock-level rates, with weights proportional to each shock's share of the first-stage covariance between sales exposure and the pre-shock forecast error, as in the Rotemberg weights of Goldsmith-Pinkham et al. (2020). A shock in which sales exposure raises the forecast error gets a negative weight. In the signed

design, at the pre-EA forecast, 13 shocks have negative weights, −0.25 in total. Over the positive-weight shocks alone, the rate is 0.32, against 0.36 for all shocks; at the final forecast, it is 0.72 against 0.69. In the unsigned design, 13 shocks have negative weights, −0.38 in total, and the rate over the positive-weight shocks is 0.24, against 0.31. The weights are concentrated. In the signed design, the five largest sum to 66 percent, and the effective number of shocks (the inverse of the sum of squared weights) is 7.6. The largest are Mexico in 2009 (0.19, with a shock-level rate of 0.03), Australia in 2020 (0.18, 0.27), China in 2018 (0.15, 0.28), Hong Kong in 2019 (0.07, 0.28) and Venezuela in 2017 (0.07, 0.20).

Table A9 estimates the signed design separately for the 24 shocks with economic measures and no war and the 15 with war on or against the shocked country's territory, and their difference in one pooled regression. For economic measures, $\pi_{\text{pre}} = 0.47$ $[0.25, 1.02]$ and $\pi_{\text{final}} = 0.81$ $[0.61, 1.09]$. For wars, the first stage has a similar slope (−2.60) but is too imprecise (effective first-stage $F = 1.6$) to identify the rate, and the revision is smaller than for economic measures (a difference of 1.59, $t = 3.16$).

A.4.4 The unsigned design's other baselines and its rate

Table A10 measures the shock year against the year before alone and against two years before alone, and the year before against the year before that, a year without a shock. Against the year before alone, the first stage is stronger than in the main specification ($t = 3.72$) and the rate is about the same ($\theta_{\text{pre}} = 0.34$ $[0.07, 0.86]$); against two years before alone, the first stage is weak ($t = 1.66$; effective first-stage $F = 2.8$). The year without a shock gives a negative slope (−1.23, $t = -1.92$), because sales-exposed firms' absolute forecast errors were unusually small in the year just before the shock. Averaging the two earlier years, as the main specification does, limits the influence of that year.

To see what the unsigned rate measures, write a firm's pre-shock forecast error as $e + s$, where s is the part that sales exposure predicts and e the rest, and suppose that revisions incorporate a share π of s, with π between 0 and 1. Measured against years without a shock, the unsigned design's first stage then reflects $g(s) = \mathrm{E}|e + s| - \mathrm{E}|e|$ and its reduced form $g(s) - g((1-\pi)s)$, so that $\theta = 1 - g((1-\pi)s)/g(s)$. Because g is convex with $g(0) = 0$, $g((1-\pi)s) \leq (1-\pi)\, g(s)$, and $\theta \geq \pi$ whenever the first stage is positive. The same bound holds when revisions also remove a share of e, as long as that share is no larger than π. Before the firm reports, that share is smaller, because the OLS coefficients at the pre-EA forecast, 0.18 and 0.20 (Tables 4 and 7), lie below the signed rate. When s is small relative to e, as it is here (Section 5.2), θ is close to $\pi(2-\pi)$. The unsigned rate therefore overstates how much analysts incorporate before the firm reports, and a set for θ_{pre} that excludes one is conservative evidence of under-reaction.

A.4.5 Revisions before the shock

Figure A3 traces how analysts' revisions vary with sales exposure, day by day, from before the shock to the pre-EA forecast, in the main specification of the signed design. It measures each revision from the analyst's forecast in effect 180 days before the shock's first month, for the 77 percent of analyst-firm-shocks that have one; Figure 3, Panel A, measures it from the pre-shock forecast. Before the shock, no day has a significant slope, and the six monthly slopes are jointly insignificant ($p = 0.57$). From 360 days before, for the 49 percent of forecasts that reach back that far, the twelve monthly slopes give $p = 0.89$. In Figure A3, the slope is -0.43 ($t = -0.75$) on the shock's first day and -1.55 ($t = -2.14$) at the pre-EA forecast. Measured against the forecast 180 days before the shock, the part of the exposure-driven shortfall already in the pre-shock forecast is 0.14 $[-0.40, 0.45]$, and counting it as incorporation would give $\pi_{\text{pre}} = 0.45$ $[0.03, 0.83]$. Placebo countries are not informative here, because exposure to large trading partners such as China, Japan and South Korea predicts downward revisions in the months before any shock. As a result, 8 of the 20 placebo countries have pre-shock slopes at least as large, in the same direction, as the shocked country's slope, which is close to zero.

A.4.6 Small sales-exposed firms and the size rule

Section 3.3 restricts the main sample to firms above the shock's median market value, because of analysts' standing optimism about small sales-exposed firms. Table A11 shows that optimism by repeating the firm-level comparison of Table 5 for the smaller firms (Panel A) and for all firms (Panel B). The regressions are those of Table 5: the I/B/E/S consensus in the shock year and in each of the two years before it, sales exposure counted in the whole 10-K with sourcing and operations exposure included, the six standard and six firm controls and industry-by-shock fixed effects, for clean, non-financial firms.

Among small firms, the pre-shock forecast error falls with sales exposure in the shock year (-11.79, $t = -2.47$), and it also falls in the year before (-7.82, $t = -1.92$) and two years before (-7.45, $t = -1.91$). Measured against the mean of the two earlier years, the shock-year slope is -4.15 ($t = -0.70$). Relative to other small firms in the same industry, analysts' forecasts for small sales-exposed firms are too high whether or not a shock occurs, consistent with optimism being larger where information is harder to obtain (Lim, 2001). The shock year is not special for these firms' forecast errors.

Pooling small and large firms gives a similar picture (Panel B). The pre-shock forecast error falls with sales exposure in the shock year (-6.57, $t = -2.63$) and in each of the two earlier years (-3.87, $t = -2.19$, and -3.30, $t = -2.56$), so the shock year differs little from

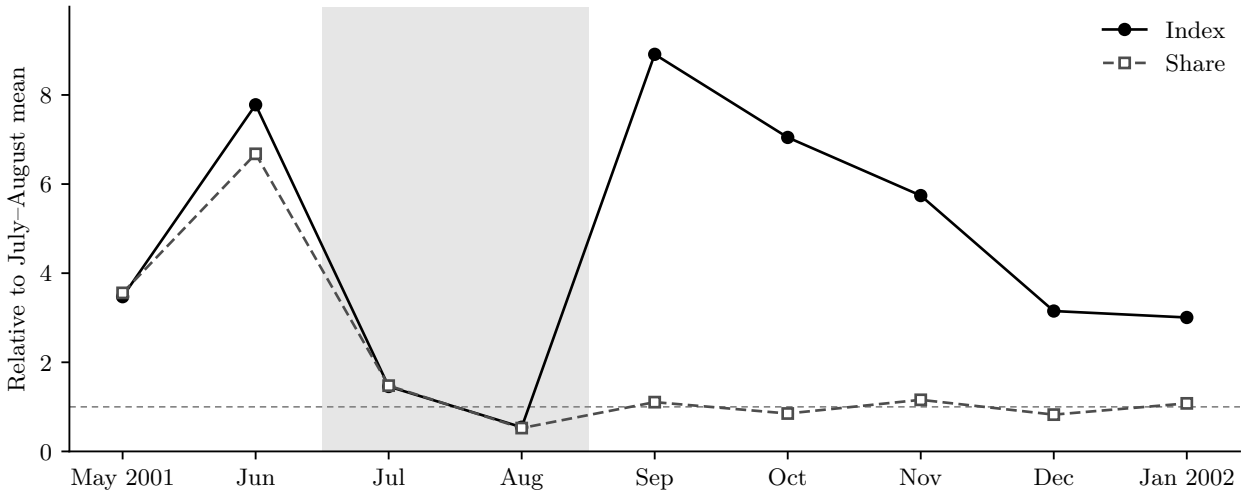
them (-2.99, $t = -1.13$). In the main sample of large, clean firms, sales exposure predicts no shortfall in either of the two years before the shock ($t = -0.04$ and -1.39; Table 5), so the shortfall there is specific to the shock year.

Among the firm-shocks that the size rule ranks (non-financial firms with the exposure measures and firm controls that pass the other filters, before the clean filter), the larger half holds 97.6 percent of market value, pooled over the 39 shocks, and between 93.7 and 98.7 percent within each shock. Its firms have a median of eight analysts, and 10 percent have one or two; the smaller half has a median of three, and 43 percent have one or two. Measured against NYSE size breakpoints in the month before each shock, the cutoff averages 62 percent of the NYSE median. In the size groups of Fama and French (2008), the larger half has no microcaps (stocks below the NYSE 20th percentile) and is 77 percent big stocks, and the smaller half has no big stocks and is 61 percent microcaps. The smaller half amounts to about 2 percent of U.S. stock market value in the median shock.

A.4.7 Visible exposure, dispersion and final forecasts

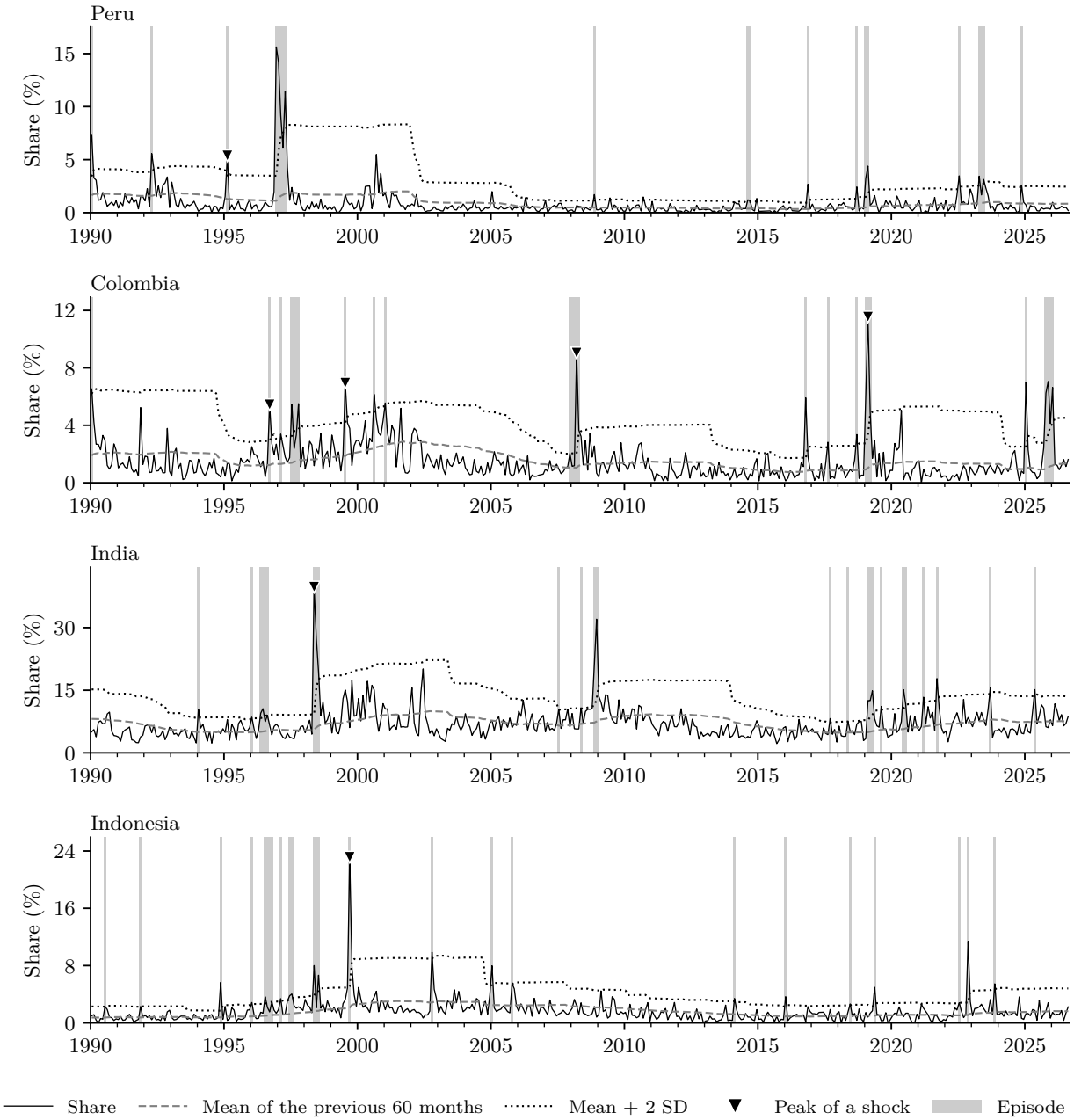
Table A12 reports the estimates behind Sections 5.3, 7.2 and 7.5. Panel A estimates the signed design separately for the sales-exposed firms whose exposure is visible as a geographic segment named for the shocked country and for the other sales-exposed firms. Panel B relates the change in forecast dispersion to sales exposure, with dispersion measured from the consensus and from the firm's own analysts. Panel C divides analysts' final forecasts by when the analyst last issued or confirmed them and compares them with the consensus.

Figure A1: From the Geopolitical Risk Index to the Share of Geopolitical-Risk News



Sweden, May 2001 to January 2002: its index, the percent of all articles in the ten newspapers that concern geopolitical risk and mention Sweden, and its share of geopolitical-risk articles, each divided by its mean over July and August 2001 (shaded). In September 2001 the index rose to 8.9 times that mean (6 and 16 times its July and August values), while the share was 1.1 times its mean (1.0 percent of geopolitical-risk articles, against 1.4 in July and 0.5 in August): world news raised the index, not Sweden's part in it. In June both rose with Sweden's own news, the Gothenburg EU summit riots. The index is that of Caldara and Iacoviello (2022); the shocks are defined on the share (Figure A2).

Figure A2: The Episode Rule in Each Shocked Country



Each panel is one of the 22 shocked countries, in the order of Figure 1: the monthly share of geopolitical-risk articles that mention the country, in percent; the mean of the previous 60 months (dashed) and that mean plus two standard deviations (dotted); the country's episodes (shaded); and the peak months of its shocks (triangles). A month is elevated when its share exceeds the threshold and at least ten geopolitical-risk articles mention the country; elevated months separated by at most two other months form one episode. The triangles mark the 39 shocks of Table 1: episodes peaking at least three standard deviations above the mean that pass the selection of Table A3. These countries have 337 episodes in all. The series are shown from January 1990, the first month the rule judges (earlier months only feed the first 60-month mean), to August 2026; each panel has its own vertical axis.

Figure A2: The Episode Rule in Each Shocked Country (continued)

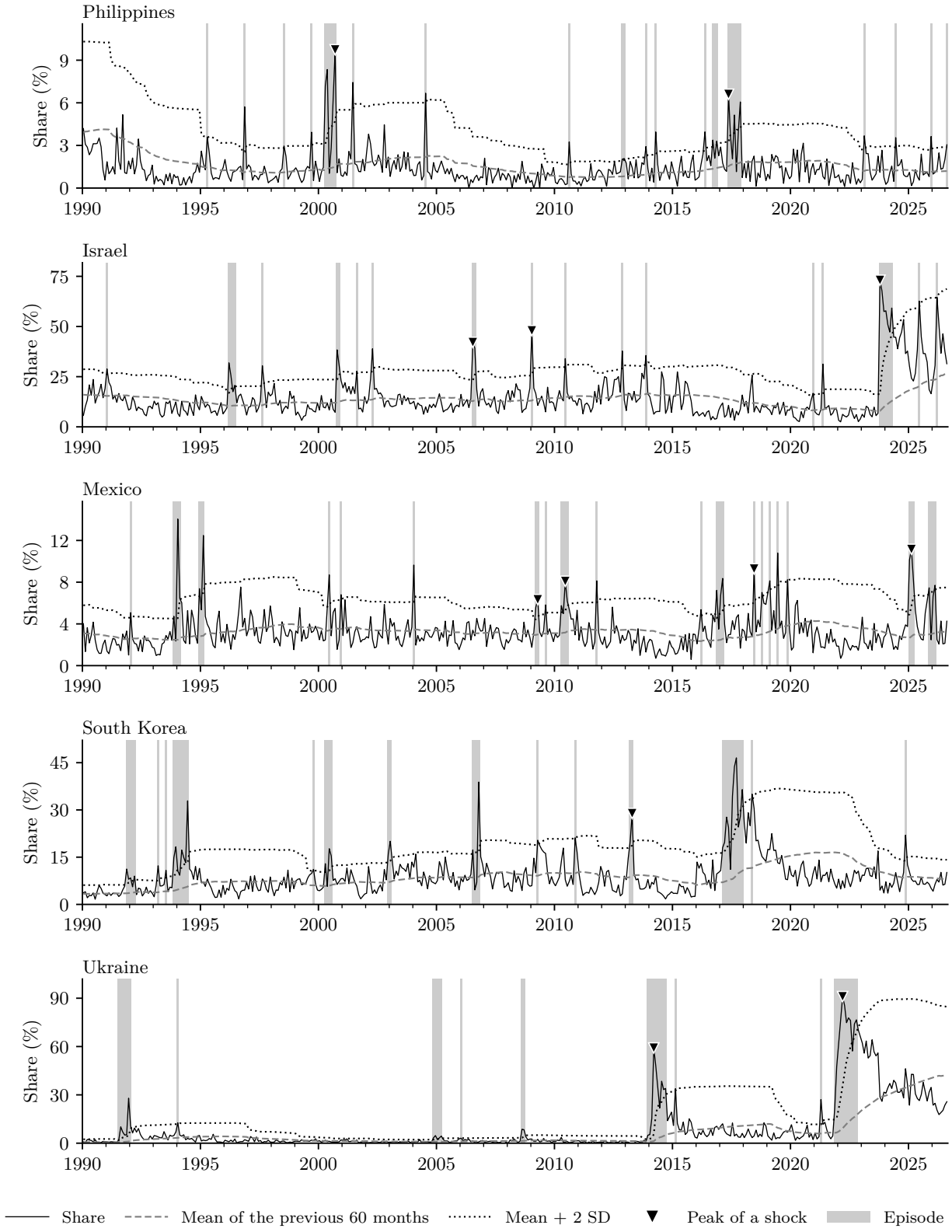


Figure A2: The Episode Rule in Each Shocked Country (continued)

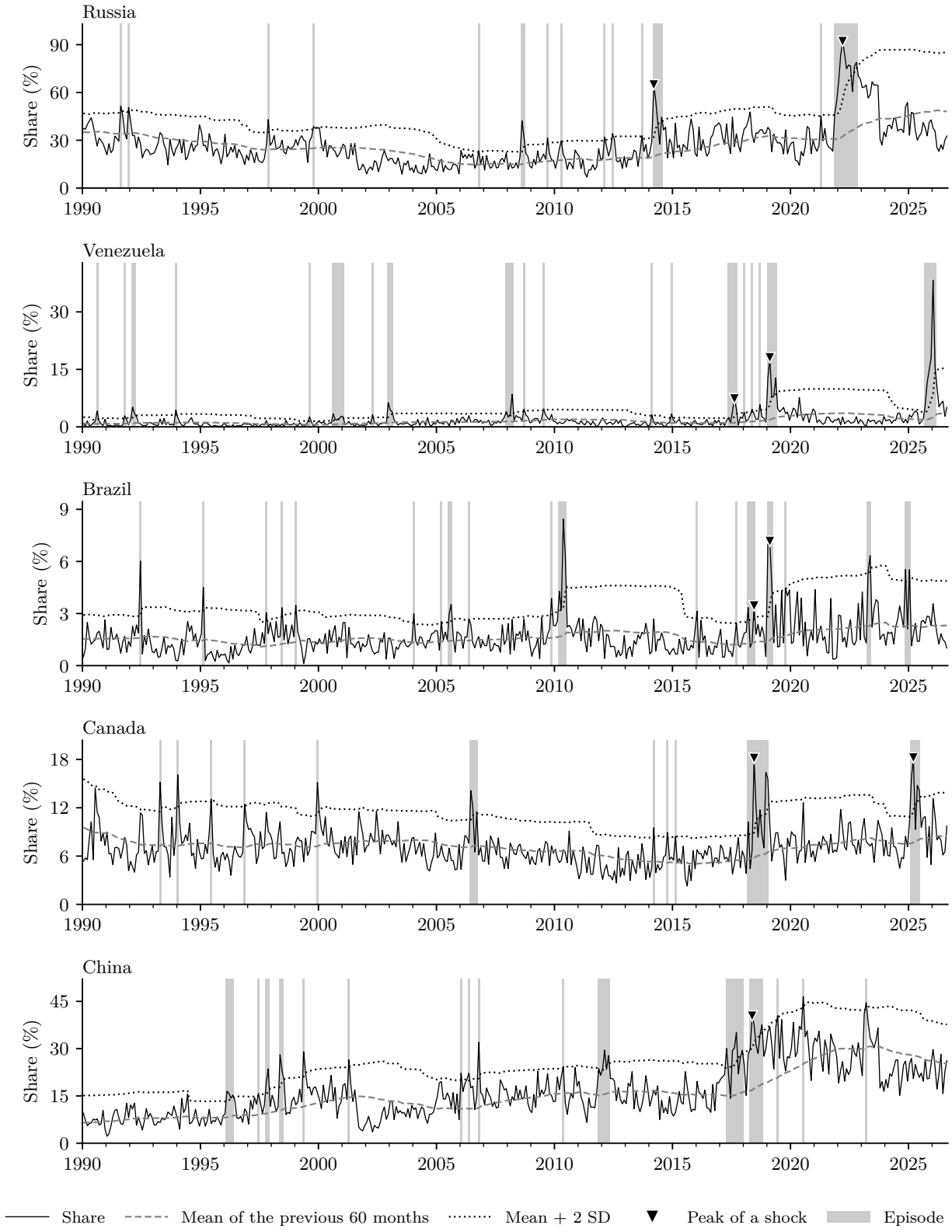


Figure A2: The Episode Rule in Each Shocked Country (continued)

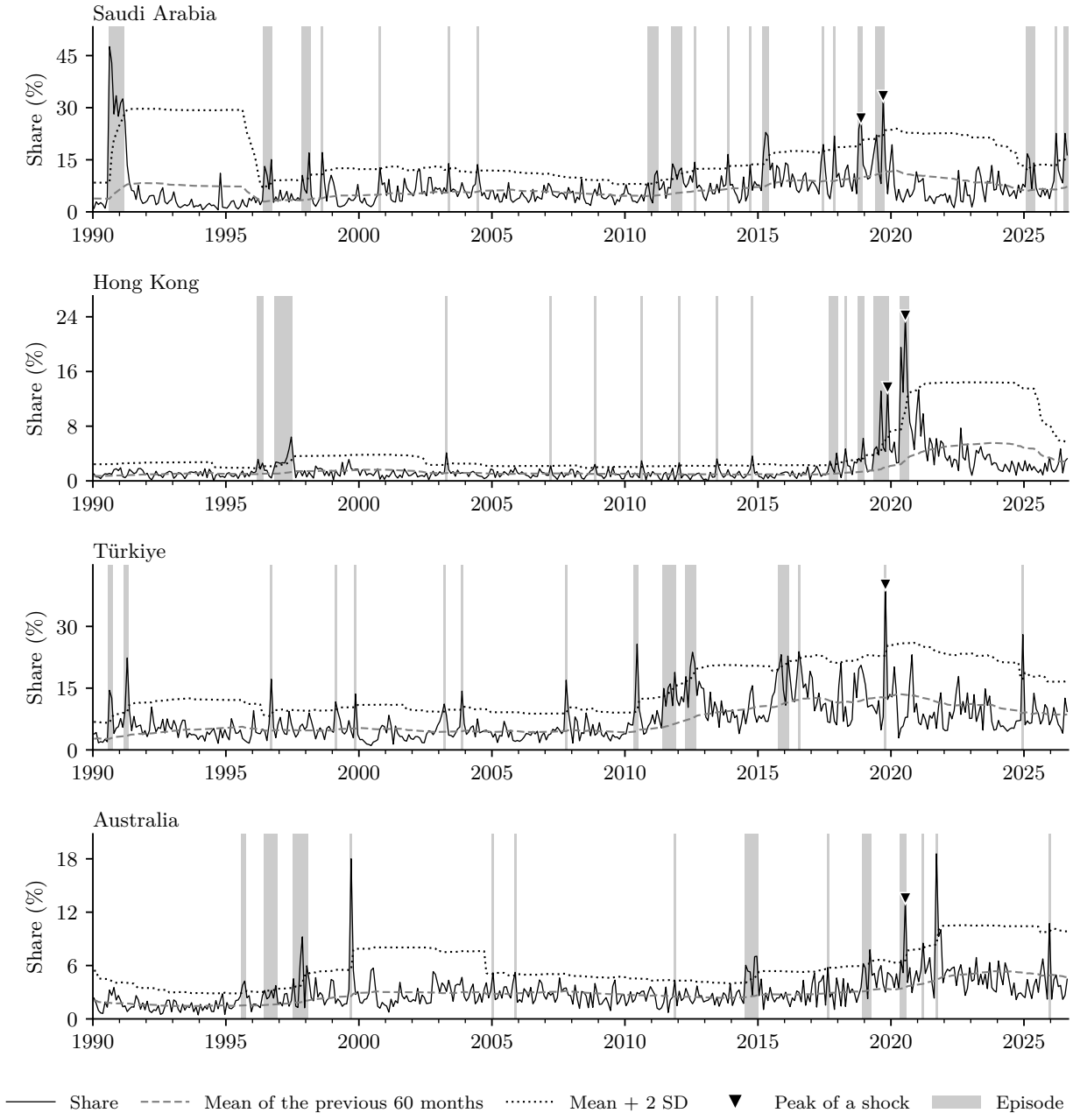


Figure A2: The Episode Rule in Each Shocked Country (continued)

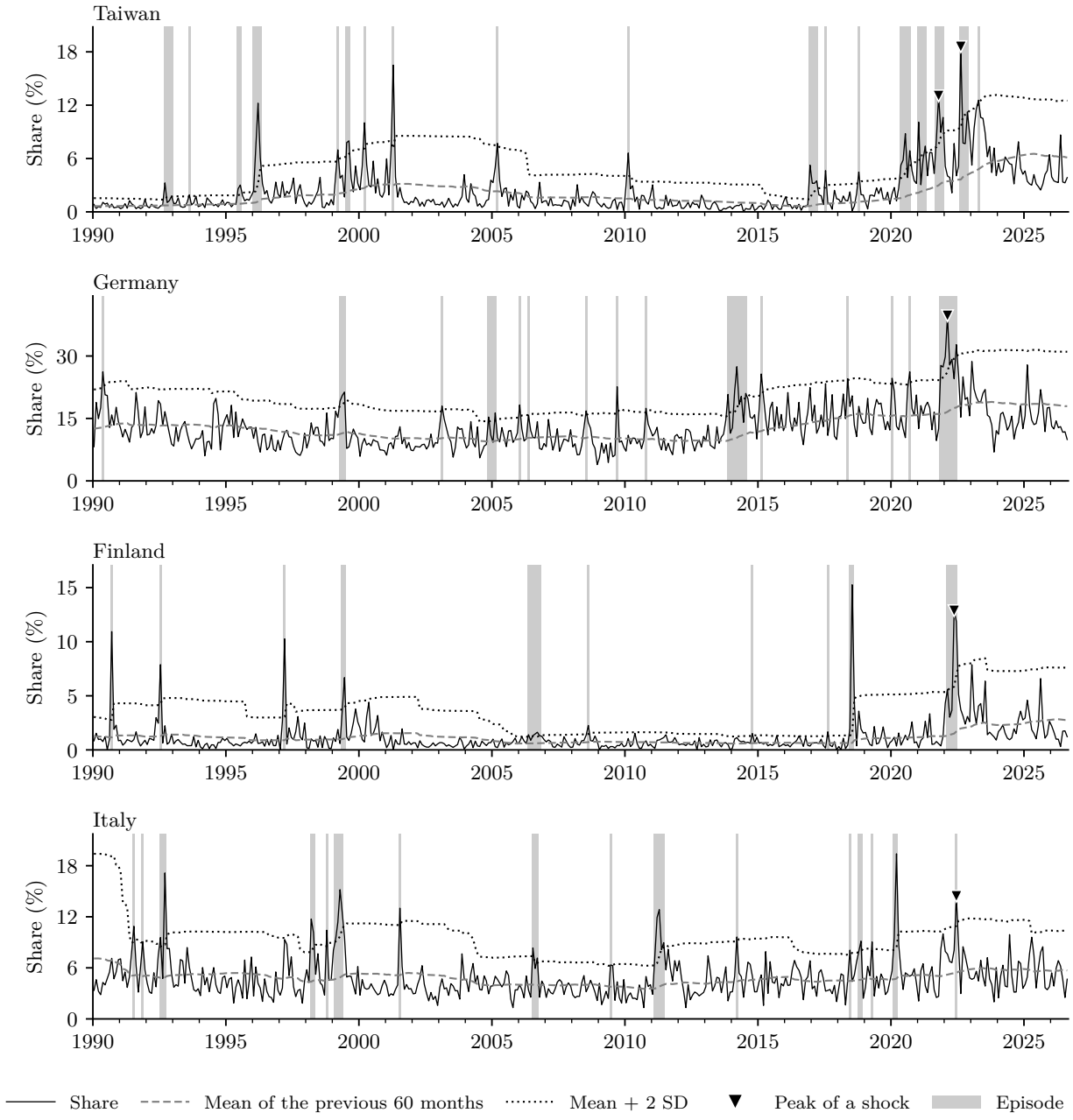
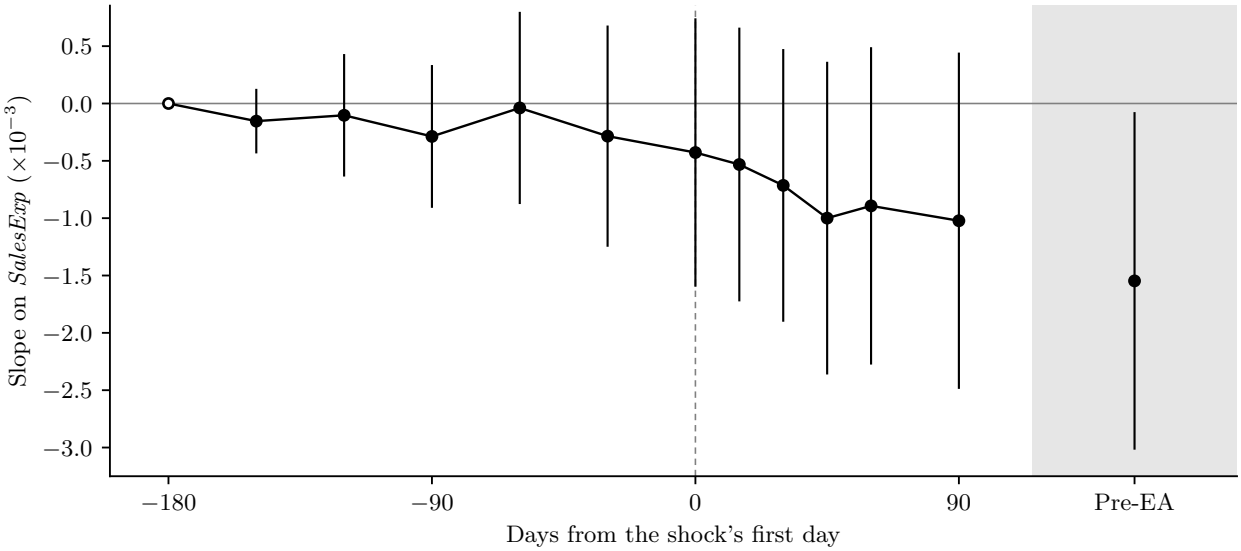


Figure A3: Revisions before and after the Shock



The slope on sales exposure of each analyst's cumulative forecast revision since the forecast in effect 180 days before the first day of the shock's first month, in thousandths of the share price, with its 95% interval, by day from the first day of the shock's first month (dashed line), and at the pre-EA forecast (right, shaded), for the 77% of analyst-firm-shocks that have a forecast on the base day. The hollow circle marks the base day, where the revision is zero by construction. The slope is -0.43 ($t = -0.75$) on the shock's first day and -1.55 ($t = -2.14$) at the pre-EA forecast; the pre-shock forecast error's slope on these rows is -2.69. Figure 3, Panel A, measures the revision from the pre-shock forecast F^0 for every analyst row instead. Each day's forecast is the analyst's latest before it. The specification is Table 4, column (3); standard errors are clustered by shock (39 clusters).

Table A1: Variable Definitions

Variable	Definition	Source
Signed design		
FE^0	Pre-shock forecast error: actual street EPS for the target year minus the analyst's pre-shock forecast, over P. The target year is the first fiscal year ending at least six months after the peak month; the pre-shock forecast is the analyst's last forecast of it issued before the shock's first month and issued or confirmed after the firm's previous earnings announcement. Winsorized at the 1st and 99th percentiles within each shock.	I/B/E/S
REV^r	Forecast revision: the analyst's post-shock forecast r minus the pre-shock forecast, over P; winsorized likewise. Pre-EA: the last forecast issued before the firm's first earnings announcement (EA) after the peak month. Post-EA: the first forecast on or after that announcement. Final: the last forecast before the target year's annual earnings announcement.	I/B/E/S
FE^r	Forecast error of forecast r: $FE^0 - REV^r$.	
P	Share price at the end of the month before the shock's first month.	CRSP
Unsigned design		
AFE^0, IMP^r	For the fiscal year t relative to the target year ($t = 0, -1, -2$), from the consensus $\bar{F}$ taken $\|t\|$ years before the shock's dates: the absolute pre-shock forecast error $\|A - \bar{F}^0\|/P$, and the accuracy improvement $AFE^0 - AFE^r$, where $AFE^r = \|A - \bar{F}^r\|/P$.	I/B/E/S, CRSP
ΔAFE, ΔIMP^r	The shock year's value minus the mean of the two earlier years'.	
Exposure, from the 10-K in use		
SalesExp	Sales exposure: $\ln(1 + x)$, where x is the shocked country's mentions within 25 words of a sales word (sales, customers, markets, exports, stores, for example) in the same paragraph, per 10,000 words. The 10-K in use is the latest filed before the shock's first month, at most 18 months old, and kept while an episode of elevated geopolitical risk in a country it mentions is under way.	10-K
SourceExp, *OpsExp*	Sourcing and operations exposure: the same with sourcing words (supplier, import, vendor, subcontractor) or with operations words (subsidiary, facility, plant, factory, warehouse; manufacture, production).	10-K
TotalExp	Total exposure: the same with every mention of the shocked country. Other mentions (Table 8) are those with no sales, sourcing or operations word nearby.	10-K
Section, placebo	The same for mentions in Item 7, Item 1 or Item 1A, per 10,000 words of that item; or for a country other than the shocked one.	10-K

Table A1: Variable Definitions (continued)

Variable	Definition	Source
Standard controls		
Log market value	Of all share classes linked to the firm's GVKEY ($ millions), at the end of the month before the shock's first month.	CRSP, CCM
Countries mentioned	Non-U.S. countries the 10-K in use mentions.	10-K
ln(1 + foreign mentions)	Mentions of non-U.S. countries per 10,000 words of the 10-K in use.	10-K
Log number of analysts	The firm's analysts in the shock.	I/B/E/S
Months, days	Months from the shock's first month to the end of the target year; days from the end of the peak month to the day before the firm's first earnings announcement after it.	I/B/E/S
Firm controls		
ROA, leverage	Income before extraordinary items, and long-term plus current debt, over total assets, for the fiscal year reported in the 10-K in use. Continuous firm controls are winsorized at the 1st and 99th percentiles.	Compustat
Book-to-market, loss	Common equity over market value; an indicator of negative income before extraordinary items. Same year.	Compustat
Earnings volatility	Standard deviation of ROA over the five fiscal years to that year (at least three).	Compustat
Past 12-month return	Compounded over the 12 months before the shock's first month (at least ten).	CRSP
Samples		
Large	Market value above the shock's median among the design's firm-shocks that pass the common filters and have the controls (the same rule in both designs; non-financial firms with the firm controls in the signed design, all industries in the unsigned design).	CRSP
Clean	No restatement or Item 4.02 non-reliance filing for the target or the previous fiscal year; special items at most 5% of assets in the target year.	Audit Analytics, Compustat
Non-financial	Fama–French 48 industries other than 44–47.	CRSP
All samples	Price at least $1; pre-EA forecast within 120 days of the end of the peak month; no acquisition above a quarter of the previous year's sales.	CRSP, Compustat
Experience		
Same-country	1 if, during an earlier, finished episode of economic measures or war that directly involved the same country (one of 102), the analyst forecast a firm whose 10-K mentioned selling there, and that firm's annual earnings were out before the new shock's first month.	I/B/E/S, 10-K
Amount	$\ln(1+n)$, standardized; n counts such earlier episodes in any country.	I/B/E/S, 10-K

I/B/E/S detail files give the analysts' forecasts and the actuals, its summary files the consensus; 10-Ks are from SEC EDGAR; market data are from CRSP and accounting data from Compustat, linked through the CRSP/Compustat Merged (CCM) links valid before the shock; restatements are from Audit Analytics.

Table A2: Sample Selection

	Firm-shocks	Analyst-firm-shocks
Panel A. Both designs		
Firms with a 10-K in use at the shock's first month	248,504	
Linked to I/B/E/S and CRSP on the day before the shock	152,999	
With a target year, its actual EPS and announcement dates	137,365	
Analysts' last forecasts of the target year's EPS before the shock		1,136,132
Issued or confirmed after the firm's last earnings announcement		737,092
Not excluded by I/B/E/S or stopped by the broker		624,099
Passing the data checks		590,531
With at least one analyst left	101,264	590,531
Panel B. Signed design: analysts' own forecasts		
Share price of at least $1	100,305	588,486
Pre-EA forecast within 120 days after the end of the peak month	100,050	587,594
No acquisition above 25% of the previous year's sales	95,178	558,762
Outside the financial industries (Fama–French 44–47)	74,127	459,242
Exposure and controls available	71,505	450,187
Large: market value above the shock's median	35,742	323,785
No restatement or non-reliance filing	31,389	287,752
No special items above 5% of assets	**29,177**	**268,865**
Panel C. Unsigned design: the consensus forecast, all industries		
Panel B's first three filters	95,178	
Large: market value above the shock's median	47,581	
No restatement or non-reliance filing	41,987	
Consensus forecasts and actual EPS for the two earlier fiscal years	**36,977**	

Observations left after each step, for street EPS in the 39 shocks; the last row of Panels B and C, in bold, is each design's sample. A firm-shock is one firm in one shock; an analyst-firm-shock is one analyst's forecasts of that firm's target-year EPS in that shock. Each firm also needs an earnings announcement before the shock and one after its peak. The indented rows count each analyst's last forecast of the target year before the shock's first month, for individual analysts only. The data checks drop forecasts whose target-year actual EPS is missing, announced before the first announcement after the peak, or in another currency, forecasts of firms without a share price, forecasts spanning a stock split or on another share basis than the actual, and the second broker of an analyst who moved. A firm-shock leaves the sample when none of its analysts is left. The acquisition filter applies to the target year. Large firms are those above the shock's median market value among the firm-shocks left at that step, so the step keeps about half of them; Panel C applies the same rule to its own firm-shocks, which include financial firms. Panel C's last step requires the same firm's consensus forecasts and actual EPS for each of the two fiscal years before the target year, taken at the same points in time one and two years earlier. Appendix Table A1 defines the filters.

Table A3: Coding Scheme and Selection of Shocks

Panel A. What is coded for each episode

Main event	The event that drove the peak month's coverage, named alike in every country.
Types	Every type of act in that event that the country took part in, and the main one.
Involvement	Direct if the country plays the role the main type names; spillover if only affected.
Role	The country's part in the main type: sender, target (or both), location or party.
Counterparts	The other end of that role: countries, organizations or non-state actors.
War	At war scale: 1,000 battle deaths in a year, or a large-scale invasion.

Panel B. Event types

Type	Definition	Based on
Armed conflict and clashes	Combat of any size between armed forces or groups, between or within states, up to war.	UCDP; MID
Terrorism	An attack by a non-state actor with a political, economic, religious or social goal, meant to intimidate a wider audience.	GTD
Use of force without combat	Force against another state's forces, officials, ships or aircraft without an exchange of fire: seizures, boardings, ramming.	MID (seizure)
Covert action	Secret, unclaimed acts against another state: targeted killings, sabotage, cyberattacks, election interference.	U.S. statute
Coups and civil unrest	Coups and self-coups; mass protests against the government; riots; revolts short of organized armed fighting.	Powell–Thyne; MMP
Economic measures	Coercive steps by states that restrict what private parties may trade or finance with another economy, or cut state-controlled energy.	GSDB; ICB
Diplomatic measures	Steps aimed at a government: expelling diplomats, severing relations, sanctions on named individuals, aid suspensions.	ICB
Threats	Signals of intent or capability not carried out within the episode: verbal, military, economic or terror threats.	GPR; MID; ICB

Panel C. From coded episodes to shocks

Step	Episodes (events)
Episodes coded (outside the United States)	651 (361)
Direct involvement	377 (232)
With economic measures or war	102 (55)
Peak at least 3 SD	66 (40)
On the receiving end and known by the peak month	41 (27)
With actual EPS available	39 (26)

An episode is a country's run of months in which its share of the Caldara–Iacoviello geopolitical-risk articles exceeds its mean over the previous 60 months by more than two standard deviations. Each label cites at least three sources. The types adapt, without their thresholds (such as UCDP's 25 deaths): UCDP/PRIO armed conflicts (Gleditsch et al., 2002) and UCDP non-state conflicts (Sundberg et al., 2012; Davies et al., 2026) (UCDP); Militarized Interstate Disputes (MID) (Palmer et al., 2022); the Global Terrorism Database (GTD) (LaFree and Dugan, 2007); the Mass Mobilization Project (MMP) (Clark and Regan, 2016); the Global Sanctions Data Base (Felbermayr et al., 2020) (GSDB), narrowed; International Crisis Behavior's economic, political and verbal acts (ICB) (Brecher and Wilkenfeld, 1997; Brecher et al., 2025); and the threat categories of Caldara and Iacoviello (2022) (GPR). Covert action follows the U.S. statutory definition (50 U.S.C. 3093(e)), O'Rourke (2018) and Carson (2018), and coups follow Powell and Thyne (2011). Panel C counts episodes, with distinct events in parentheses. Table A4 lists the candidates that the receiving-end and known-by-the-peak rules exclude, and the 2 set aside for want of actual EPS.

Table A4: Candidate Shocks Left Out

Country	Peak	Event	Reason
Saudi Arabia	Aug 1990	Iraqi invasion of Kuwait and the Gulf War, 1990–1991	Known by the peak: fighting only from 1991
Japan	Sep 1990	Iraqi invasion of Kuwait and the Gulf War, 1990–1991	Sender only: its own sanctions on Iraq
Spain	May 1993	Bosnian War, 1992–1995	Peacekeepers abroad
Netherlands	Jul 1995	Bosnian War, 1992–1995	Peacekeepers abroad
Belgium	Apr 1999	Kosovo War, 1998–1999	Air campaign abroad
Italy	Apr 1999	Kosovo War, 1998–1999	Air campaign abroad
China	May 1999	Kosovo War, 1998–1999	Embassy bombed abroad; no economic hit
Venezuela	Apr 2002	Venezuelan coup attempt, 2002	Other event: own measure; a coup
Japan	Oct 2006	North Korean missile and nuclear tests, 2006	Sanctions were its own; threats only
Venezuela	Mar 2008	Andean diplomatic crisis, 2008	Sender only: its border closure
Russia	Aug 2008	Russo-Georgian War, 2008	Fought abroad; no major measures on it
Egypt	Jan 2009	Gaza War, 2008–2009	Other event: war in Gaza; stray fire
Germany	Sep 2009	War in Afghanistan, 2001–2021	War abroad
Argentina	Feb 2010	Falklands oil exploration dispute, 2010	Sender only: its shipping rule
France	Mar 2011	Libyan civil war, 2011	Air strikes abroad
Belgium	Apr 2011	Libyan civil war, 2011	Air campaign abroad
Italy	Apr 2011	Libyan civil war, 2011	Air campaign abroad
Germany	Mar 2014	Russo-Ukrainian War, 2014–	Sender only: its warning to Russia
Saudi Arabia	Apr 2015	Yemeni civil war, 2014–	Strikes abroad; small border fighting
France	Nov 2015	Paris attacks, 2015	Other event: war abroad; terrorism
India	Apr 2019	India–Pakistan border skirmishes, 2019	Other event: own measures; attack, clash
India	Jun 2020	China–India border clashes, 2020–2021	Other event: own app bans; border clash
Egypt	Oct 2023	Gaza War, 2023–2025	Other event: war in Gaza; stray strikes
Türkiye	Dec 2024	Syrian civil war, 2011–2024	Forces fighting in Syria
Venezuela	Jan 2026	U.S. military campaign in the Caribbean and Venezuela, 2025–2026	No measure against it; no war
Saudi Arabia	Mar 2026	Iran war, 2026–	Set aside: no actual EPS until 2027
Saudi Arabia	Jul 2026	Iran war, 2026–	Set aside: no actual EPS yet

The 27 candidates that pass the first four steps of Table A3, Panel C, but are not shocks (Table 1 lists the shocks), by peak month. Receiving end: the hit must come through what qualifies the episode, economic measures against the country or war on or against its territory; countries that only imposed measures (sender) or fought abroad are out, as are hits through a different, smaller event such as a terrorist attack or a border clash (other event). Known by the peak: the qualifying measures or war must be known by the end of the peak month. The last 2 pass both rules but are set aside until their target years have actual EPS.

Table A5: The Country Dictionary: What Counts and What Does Not

Panel A. Terms that count, for nine countries

Country	Names	Demonyms	Abbreviations
China	China, People's Republic of China	Chinese[c]	P.R.C., PRC
Russia	Russia, Russian Federation	Russian, Russians	—
Mexico	Mexico, United Mexican States	Mexican, Mexicans	—
Canada	Canada	Canadian, Canadians	—
Ukraine	Ukraine	Ukrainian, Ukrainians	—
Israel	Israel, State of Israel	Israeli, Israelis	—
South Korea	South Korea, Republic of Korea, Korea[a]	South Korean, Korean[a], Koreans[a]	R.O.K., ROK
Taiwan	Taiwan, Chinese Taipei[a], Republic of China[a]	Taiwanese	—
United Kingdom	United Kingdom, United Kingdom of Great Britain and Northern Ireland, Wales[b], England[b], Scotland[b]	British[c], Briton, Britons	U.K., UK

Panel B. What does not count

Rule	Examples	Why
Lowercase forms	"turkey", "china"	Only capitalized forms count.
U.S. terms	"United States", "U.S.", "American"	Domestic mentions are not counted.
Names of U.S. states	"Georgia"	A state's name counts for no country.
Places that contain a country's name	"New Mexico", "New Jersey", "New England", "British Columbia", "New South Wales"	The place name blocks the term inside it.
Seas, gulfs and straits	"Gulf of Mexico", "South China Sea", "Taiwan Strait", "Sea of Japan"	The marine name blocks the term inside it.
U.S. addresses	"Lebanon, PA", "Peru, Illinois"	A name followed by a U.S. state is a U.S. town.
Terms shared by several countries	"Congo", "Dominican", "Virgin Islands"	No single sovereign state among them.
The filer's own name	—	A company named after a country is not a mention of it.
Terms excluded by the audit	"Jordan", "Jersey", "Chad", "Turk", "Monaco", "Dane", "Finn", "CH", "NL", "PNG"	Mostly not the country in a random sample (15 terms).
Phrases blocked by the audit	"British thermal unit", "Dutch auction", "Indian tribe", "Inner Mongolia", "Uruguay Round", "Italian restaurant"	A kept term not meaning the country (24 phrases).

Terms of the rule-based country dictionary (1,162 terms for 248 countries and territories, 86 blocking phrases), after the term audit's exclusions and blocked phrases (Appendix A.3). Panel A lists every counted term of nine countries, except spelling variants ("Peoples Republic of China", "P. R. C."). [a] Stated exception. [b] Constituent country, counted for its parent. [c] Shared term, counted for the only sovereign state among the countries the sources give it to. Blocking phrases also cover their hyphenated forms. The full dictionary and the audit's lists are in the replication files.

Table A6: Further Examples of Country Mentions

Group	Firm, shock and source	Passage
Sales	PepsiCo (Russia, 2014) 10-K of Feb. 21, 2013 Item 7 (MD&A)	Our operations outside of the U.S. generate 49% of our net *revenue*, with **Russia**, Mexico, Canada, the United Kingdom and Brazil comprising approximately 25% of our net revenue.
	Procter & Gamble (Canada, 2025) 10-K of Aug. 5, 2024 Item 7 (MD&A)	Our largest international *markets* are Greater China, the United Kingdom, **Canada**, Japan and Germany and collectively comprised approximately 20% of our net sales in fiscal 2024.
Sourcing	Texas Roadhouse (Canada, 2025) 10-K of Feb. 23, 2024 Item 1A (Risk Factors)	We currently purchase our beef primarily from four beef *suppliers* coming from the United States or **Canada**.
	U.S. Steel (Russia, 2022) 10-K of Feb. 12, 2021 Item 1 (Business)	Substantially all of USSE's iron ore requirements are *purchased* from outside sources, primarily **Russian** and Ukrainian mining companies.
Operations	Chevron (China, 2018) 10-K of Feb. 22, 2018 Item 1 (Business)	In 2017, design work continued for a planned manufacturing *plant* in Ningbo, **China**, with a final investment decision expected in 2018.
	ExxonMobil (Russia, 2014) 10-K of Feb. 27, 2013 Item 2 (Properties)	ExxonMobil and Rosneft signed a Strategic Cooperation Agreement in 2011 to jointly participate in *exploration* and development activities in **Russia**, the United States and other parts of the world.
Sales and operations	Coca-Cola (Mexico, 2018) 10-K of Feb. 23, 2018 Item 1 (Business)	In **Mexico** and Brazil, we *manufacture*, *market* and sell Del Valle beverage products through joint ventures with our bottling partners.
Not counted	Philip Morris International (Canada, 2018) 10-K of Feb. 13, 2018 Item 1A (Risk Factors)	Damages claimed in some tobacco-related litigation are significant and, in certain cases in Brazil, **Canada** and Nigeria, range into the billions of U.S. dollars.
	Facebook (China, 2018) 10-K of Feb. 1, 2018 Item 1A (Risk Factors)	For example, user access to Facebook and certain of our other products has been or is currently restricted in whole or in part in **China**, Iran, and North Korea.

Each row quotes one mention of the shocked country in the 10-K in use (Section 3.1) of a firm in the signed design's main sample for that shock. Groups, bold and italics are as in Table 2. Quotes are verbatim from the 10-K, at most 35 words. Examples come from well-known firms and shocks, with a different firm in each row. Each mention belongs to its row's group or groups only.

Table A7: Placebo Countries

	Signed design (main)			Unsigned design	
Country	FE^0	REV^{pre}	FE^{pre}	ΔAFE	ΔIMP^{pre}
The shocked country	−3.49	−3.67	−2.39	3.10	2.43
Australia	0.60	1.56	0.07	0.12	−0.69
Brazil	−0.50	−1.43	−0.03	0.39	0.58
Canada	−1.71	−1.78	−1.12	0.73	0.11
China	−2.37	−1.38	−2.10	1.62	−0.19
France	−0.19	0.80	−0.39	2.41	**2.53**
Germany	1.87	0.79	1.57	0.12	0.37
Hong Kong	−1.42	−0.88	−1.41	0.40	1.32
India	0.35	−0.47	0.44	0.07	1.18
Israel	−1.44	−1.78	−0.66	−0.86	1.24
Italy	−0.08	−0.44	0.04	0.23	0.97
Japan	−0.17	−0.19	−0.14	−0.17	−0.57
Mexico	−0.01	−0.82	0.19	0.58	2.14
Netherlands	4.60†	1.19	5.17†	−1.56	−1.12
Russia	−0.15	−0.66	0.18	1.11	0.94
Singapore	2.70	2.26	2.46†	−0.84	−1.50
South Korea	0.07	0.03	0.06	−1.69	−2.86†
Spain	3.02	2.25	2.51†	0.13	−0.65
Switzerland	2.69	2.65	1.87	−1.43	−1.16
Taiwan	−0.68	−1.55	−0.07	−1.77	−1.20
United Kingdom	−0.19	−1.13	0.23	−0.27	0.52

t-statistics of the *SalesExp* slopes of Table 6, with the shocked country's mentions replaced by each placebo country's, counted the same way, and that country's own shocks dropped. Bold: as strong as the shocked country in the same direction (the same sign and at least the same absolute t-statistic); †: as strong in the opposite direction.

Table A8: Leaving Out Shocks and Clustering by Overlap Group

	Signed design (main): π_r		Unsigned design: θ_r	
	Pre-EA forecast	Final forecast	Pre-EA forecast	Final forecast
Panel A. Leaving out one shock (39 runs)				
Main estimate	0.36	0.69	0.31	0.91
	[0.17, 0.79]	[0.43, 0.93]	[0.05, 0.93]	[0.59, 1.07]
Lowest, without	0.30	0.64	0.25	0.88
	Canada 2018	Canada 2018	Canada 2025	Colombia 1996
Highest, without	0.44	0.73	0.40	0.94
	Mexico 2009	Colombia 1996	Mexico 2010	Brazil 2019
Sets excluding 0 (of 39)	39	39	39	39
Sets excluding 1 (of 39)	38	39	28	0
Effective first-stage F, range	9.33–13.62	9.33–13.62	8.01–12.17	7.42–11.74
Panel B. Leaving out one overlap group (19 runs)				
Range over the 19 runs	0.25–0.44	0.65–0.73	0.25–0.40	0.88–0.98
Without 2018–2019 (11 shocks)	0.25	0.67	0.26	0.98
	[0.05, 0.67]	[0.33, 0.93]	[0.01, 0.81]	[0.72, 1.13]
Sets excluding 0 (of 19)	19	19	18	19
Sets excluding 1 (of 19)	19	19	11	0
Panel C. Clusters by shock and by overlap group, pre-EA forecast				
	Signed design (main)		Unsigned design	
	By shock ($G = 39$)	By group ($G = 19$)	By shock ($G = 39$)	By group ($G = 19$)
FE^0 / ΔAFE	-2.42^{***}	-2.42^{***}	1.93^{***}	1.93^{**}
	(-3.49)	(-3.22)	(3.10)	(2.35)
REV^{pre} / ΔIMP^{pre}	-0.87^{***}	-0.87^{***}	0.59^{**}	0.59^{**}
	(-3.67)	(-3.64)	(2.43)	(2.31)
π_{pre} / θ_{pre}	0.36	0.36	0.31	0.31
	[0.17, 0.79]	[0.16, 0.93]	[0.05, 0.93]	[0.04, 2.11]
Effective first-stage F	12.21	10.36	9.61	5.54

Robustness of the incorporation rates of Table 4, column (3) (π_r), and Table 7, column (2) (θ_r), at the pre-EA and final forecasts. Panel A drops each of the 39 shocks in turn (38 clusters by shock in each run), with 95% Anderson–Rubin sets in brackets. Panel B drops the shocks of each of the 19 overlap groups together, with clusters by shock; 13 groups are single shocks. Overlap groups chain the shocks whose spans, from the first to the last elevated month, overlap; they are the alternative clusters for shocks that share dates. Panel C keeps every shock and clusters the standard errors by the 19 overlap groups ("by group") instead of the 39 shocks, with Anderson–Rubin sets from the same covariance. Slopes on sales exposure are in thousandths of the share price, with t-statistics in parentheses. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G - 1$ degrees of freedom; G is 39 for clusters by shock and 19 for clusters by overlap group. Whether a firm is large, the winsorizing and the weights are all determined within each shock, so dropping shocks leaves the other shocks' data unchanged. Shocks are named by country and peak year (Table 1).

Table A9: Shock Types

	Economic measures without war (1)	Wars (2)	Difference, wars minus others (3)
First stage: FE^0	-2.33^{***}	-2.60	0.68
	(-3.34)	(-1.28)	(0.31)
Reduced form: REV^{pre}	-1.10^{***}	-0.08	1.59^{***}
	(-3.69)	(-0.20)	(3.16)
IV estimate: π_{pre}	0.47	0.03	
	$[0.25, 1.02]$	$(-\infty, \infty)$	
IV estimate: π_{final}	0.81	0.32	
	$[0.61, 1.09]$	$(-\infty, \infty)$	
Effective first-stage F	11.12	1.63	
Shocks	24	15	
Firm-shocks	18,098	11,079	

Table 4, column (3), estimated separately for shocks with economic measures only (1) and for shocks with war on or against the shocked country's territory (2). Column (3) is the difference from one pooled regression that interacts each exposure variable with a war indicator; the controls' slopes are common to both types, so column (3) need not equal column (2) minus column (1). The table shows the slopes on *SalesExp*, the instrument; *SourceExp* and *OpsExp* are in every regression (and interacted with the war indicator in column (3)). Slopes are in thousandths of the share price, with t-statistics in parentheses (standard errors clustered by shock); *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G-1$ degrees of freedom; G is 24, 15 and 39 in columns (1), (2) and (3). The IV estimate π_r has its 95% Anderson–Rubin set in brackets. An Anderson–Rubin set with an infinite end is printed with a parenthesis at that end.

Table A10: The Unsigned Design against Other Baselines

	Shock year minus the year before (1)	Shock year minus two years before (2)	Year before minus two years before (3)
First stage: ΔAFE	2.60^{***}	1.21	-1.23^{*}
	(3.72)	(1.66)	(-1.92)
Reduced form: ΔIMP^{pre}	0.88^{**}	0.13	-0.83^{**}
	(2.57)	(0.49)	(-2.34)
IV estimate: θ_{pre}	0.34	0.11	0.67
	$[0.07, 0.86]$	$(-\infty, \infty)$	$(-\infty, -8.13] \cup [0.13, \infty)$
Effective first-stage F	13.83	2.76	3.69
Firm-shocks	39,788	37,723	36,977

Table 7, column (2), at the pre-EA forecast, with other baselines: ΔAFE and ΔIMP^{pre} are the shock year minus the year before (1) or minus two years before (2), or the year before minus the one before it, a year without a shock (3); the main specification subtracts the mean of the two earlier years. Variables, controls, fixed effects, weights, clustering and sample rules are those of Table 7, and each column needs consensus forecasts only in the years it uses. Slopes are in thousandths of the share price, with t-statistics in parentheses. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G-1$ degrees of freedom ($G = 39$ shocks). θ_{pre} has its 95% Anderson–Rubin set in brackets. An Anderson–Rubin set with an infinite end is printed with a parenthesis at that end.

Table A11: Small Sales-Exposed Firms

	Shock year	Year before	Two years before	Shock year minus the mean of the two years before
Panel A. Small firms (market value at or below the shock's median)				
FE^0	-11.79^{**}	-7.82^{*}	-7.45^{*}	-4.15
	(-2.47)	(-1.92)	(-1.91)	(-0.70)
Firm-shocks	20,981			
Panel B. All firms				
FE^0	-6.57^{**}	-3.87^{**}	-3.30^{**}	-2.99
	(-2.63)	(-2.19)	(-2.56)	(-1.13)
Firm-shocks	46,975			

Table 5's firm-level regressions of the pre-shock forecast error FE^0, from the I/B/E/S consensus for the shock year and each of the two fiscal years before it, for small firms (Panel A) and for all firms (Panel B) instead of large ones: clean, non-financial firm-shocks with consensus forecasts and actuals in all three years. Each regression gives the slope on sales exposure, with sourcing and operations exposure, the six standard and six firm controls, and industry × shock fixed effects. Slopes are in thousandths of the share price. The t-statistics in parentheses use standard errors clustered by shock. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G-1$ degrees of freedom ($G = 39$ shocks).

Table A12: Visible Exposure, Forecast Dispersion and Final Forecasts

Panel A. Visible and other sales exposure

	Visible exposure (1)	Other sales-exposed firms (2)	Difference (3)
IV estimate: π_{pre}	0.21	0.37	−0.16
	[−0.14, 0.42]	[0.12, 1.17]	[−0.47, 0.14]
IV estimate: π_{post}	0.41	0.44	−0.03
	[−0.03, 1.69]	[0.10, 1.41]	[−0.41, 0.35]
IV estimate: π_{final}	0.61	0.68	−0.07
	[−0.71, 0.78]	[0.32, 1.07]	[−0.39, 0.25]
Effective first-stage F, pre-EA	5.07	9.02	
Sales-exposed firm-shocks	551	4,260	

Panel B. The change in forecast dispersion

	Pre-EA forecast	Post-EA forecast	Final forecast
I/B/E/S consensus standard deviation	−0.04	0.10	0.06
	(−0.35)	(0.73)	(0.26)
Standard deviation across the firm's analysts	−0.02	0.19	0.24
	(−0.14)	(0.98)	(0.94)
Firm-shocks (first row)	24,841	24,722	24,391
Firm-shocks (second row)	26,483	25,858	26,483

Panel C. Final forecasts by the analyst's last activity

	All final forecasts (1)	Active in the last 120 days (2)	Inactive for longer (3)	Consensus (4)
IV estimate: π_{final}	0.69	0.86	0.05	0.92
	[0.43, 0.93]	[0.68, 1.18]	[−1.43, 0.48]	[0.81, 1.11]
Effective first-stage F	12.21	9.92	7.16	18.14
Share of final forecasts (%)		82.1	17.9	
Observations	268,864	220,613	48,251	25,978

Panel A: Table 4, column (3), estimated twice, once with the sales-exposed firm-shocks whose firm reports a geographic segment named for the shocked country in the fiscal year of its 10-K in use (Compustat historical segments, as first reported; column (1)) and once with the other sales-exposed firm-shocks (column (2)), each time together with the firm-shocks without sales exposure. Column (3) is their difference, with its 95 percent confidence interval in brackets. Panel B: slopes on *SalesExp* of the change in forecast dispersion from the pre-shock forecast to each post-shock forecast, in thousandths of the share price, with Table 4's column (3) regressors at the firm level. The first row measures dispersion as the I/B/E/S consensus standard deviation, for the firms of Table 5 with at least three estimates; the second as the standard deviation of the same analysts' own forecasts at the pre-shock and the post-shock forecast, for firms with at least three such analysts. For comparison, the pre-shock forecast error falls by 3.35 thousandths per unit of *SalesExp* in Table 5, and the median dispersion of sales-exposed firms before the shock, in the pre-EA sample, is 2.88 thousandths. Panel C: Table 4, column (3), at the final forecast, for every analyst's final forecast (1); for those last issued or confirmed (an I/B/E/S review date) in the 120 days before the annual announcement (2); for the others (3); and Table 5's consensus (4). Columns (2) and (3) keep the main sample's weights. Observations are analyst-firm-shocks in columns (1) to (3) and firm-shocks in column (4). The t-statistics in parentheses use standard errors clustered by shock. *, ** and *** denote two-sided significance at the 10, 5 and 1 percent levels, from the clustered t-statistic with $G-1$ degrees of freedom ($G = 39$ shocks). The IV estimate π_r has its 95% Anderson–Rubin set in brackets.

Declaration of Generative AI Use

Generative AI tools were used to assist with language editing; checking for formatting inconsistencies throughout the manuscript; preparing and formatting tables and figures; and generating code to implement regression specifications determined by the author.